RAY TRACING IN RELATIVISTIC STARS

From geometry to *observation*



How spacetime curvature decides the *image*, the *spectrum* and the *light curve* that an X-ray telescope receives from a neutron star.

$$\psi(b) = \int du \,/\, \sqrt{(b^{-2} - u^2(1 - R_s u))}$$ THE LIGHT LEAVING THE SURFACE

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OUTLINE

Three parts, *one path*

I

Relativistic ray tracing

Null geodesics in Schwarzschild: light bending, redshift and time delays — and how to follow the rays backwards, from the telescope to the surface.

II

Pulsars and neutron stars

The laboratories: structure and equation of state, pulsars and magnetars, hot spots, rotation, atmospheres — and what NICER has already measured.

III

Recent results

My published work: magnetar light curves, a massive low-field magnetar, accretion columns — and tests of alternative theories of gravity.

THE PREMISE

Everything we know arrives *as photons*

Compact stars are *natural laboratories* of strong-field gravity — but we know them only through the photons that left them and crossed a curved spacetime.

metric → null geodesics → image · spectrum · light curve → telescope → **M, R, field, gravity**

THE DIRECT PROBLEM

Given the star and the metric, predict what is observed. That is **ray tracing**.

THE INVERSE PROBLEM

Given what is observed, infer mass, radius, spot geometry and magnetic field.

THE TEST

If the metric is not Schwarzschild, the light curves change: observation tests **gravity**.

I

PART ONE

Relativistic *ray tracing*

Near a mass, light does not travel in a straight line: it follows the null geodesics of the metric. Ray tracing means following those geodesics backwards — from the telescope to the source.

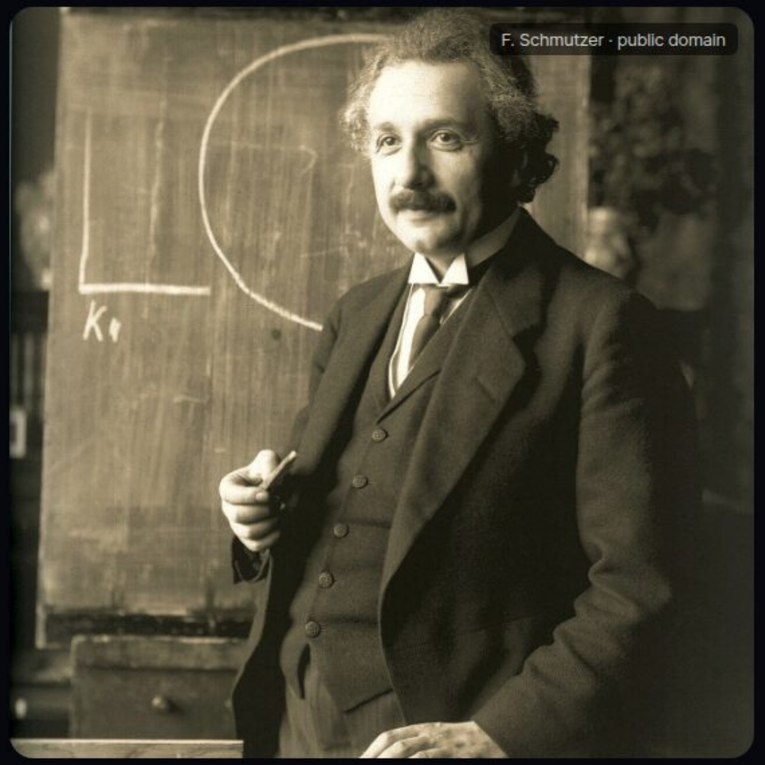
null geodesics · light bending · redshift · the image of a compact star · the recipe



SMACS 0723 (JWST) · NASA, ESA, CSA, STScI · public domain

25 NOVEMBER 1915

Einstein and *his field equations*



Albert Einstein
1921

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

CURVATURE = ENERGY AND MOMENTUM CONTENT

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu{}_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0$$

GEODESIC

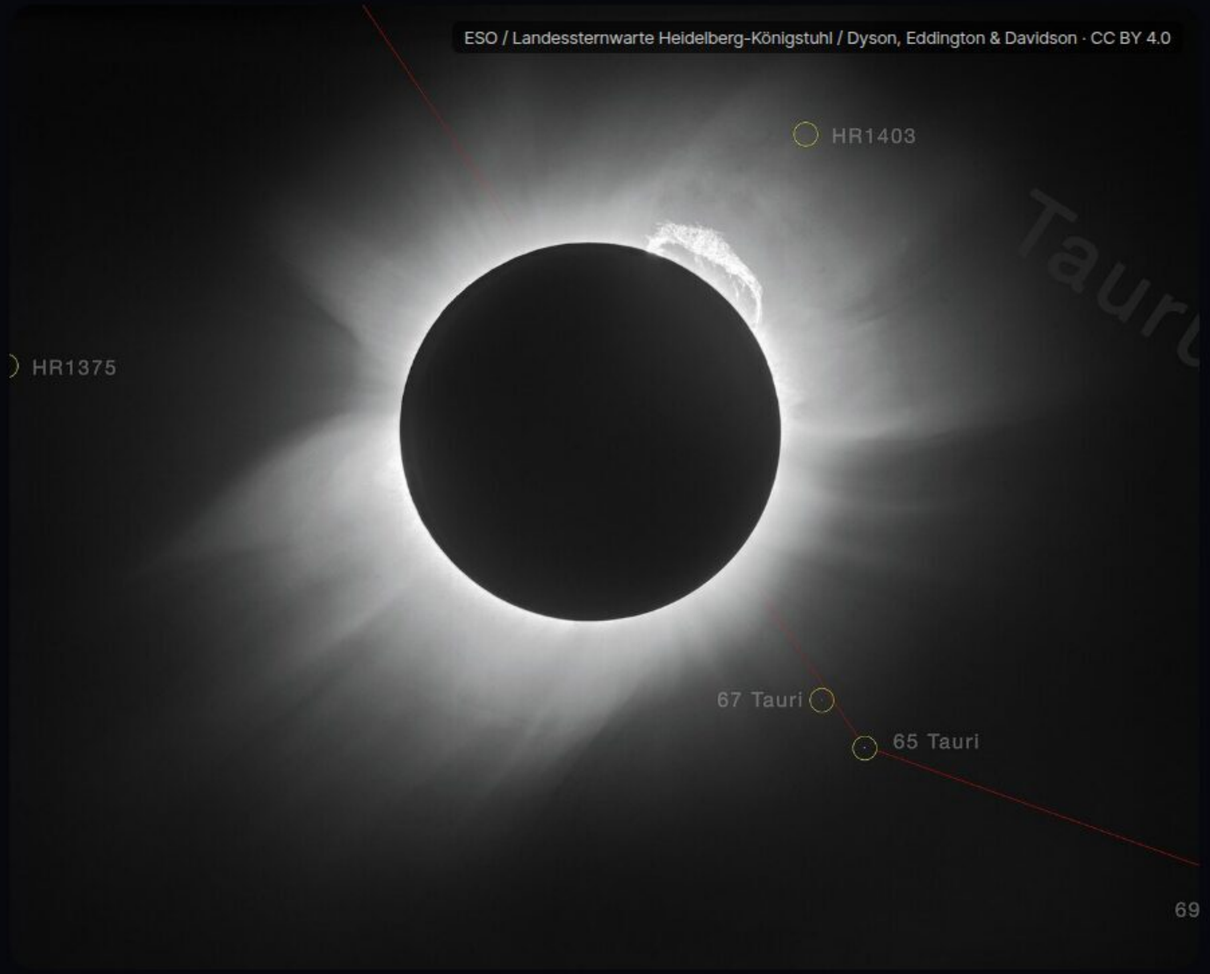
$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

THE METRIC IS THE FIELD

> *"Spacetime tells matter how to move; matter tells spacetime how to curve."*
>
> John A. Wheeler

THE CLASSICAL TESTS

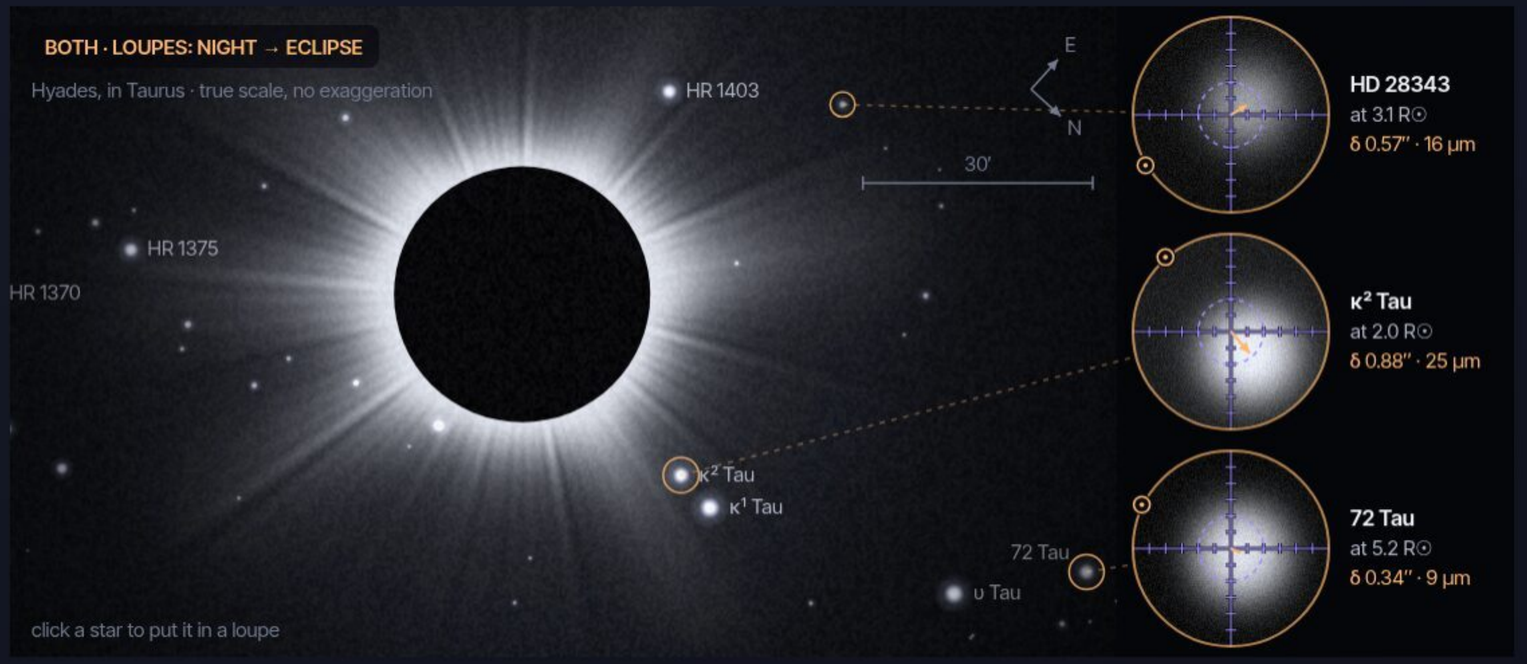
Sobral, 1919: light *bends*



EFFECT	GR PREDICTS
perihelion of Mercury (1915)	43″ / century
light grazing the Sun (1919)	1.75″
redshift, Pound–Rebka (1959)	2.5×10^{-15} over 22.5 m
Shapiro delay (1964)	~200 µs
GPS clocks	+38 µs / day

Sobral expedition (Ceará, Brazil): **1.98 ± 0.12″**. Príncipe: 1.61 ± 0.30″. The sky over north-eastern Brazil confirmed Einstein.

Two plates, *the same sky*



Night: the Hyades photographed in July, before dawn, with the same telescope — the Sun far away.

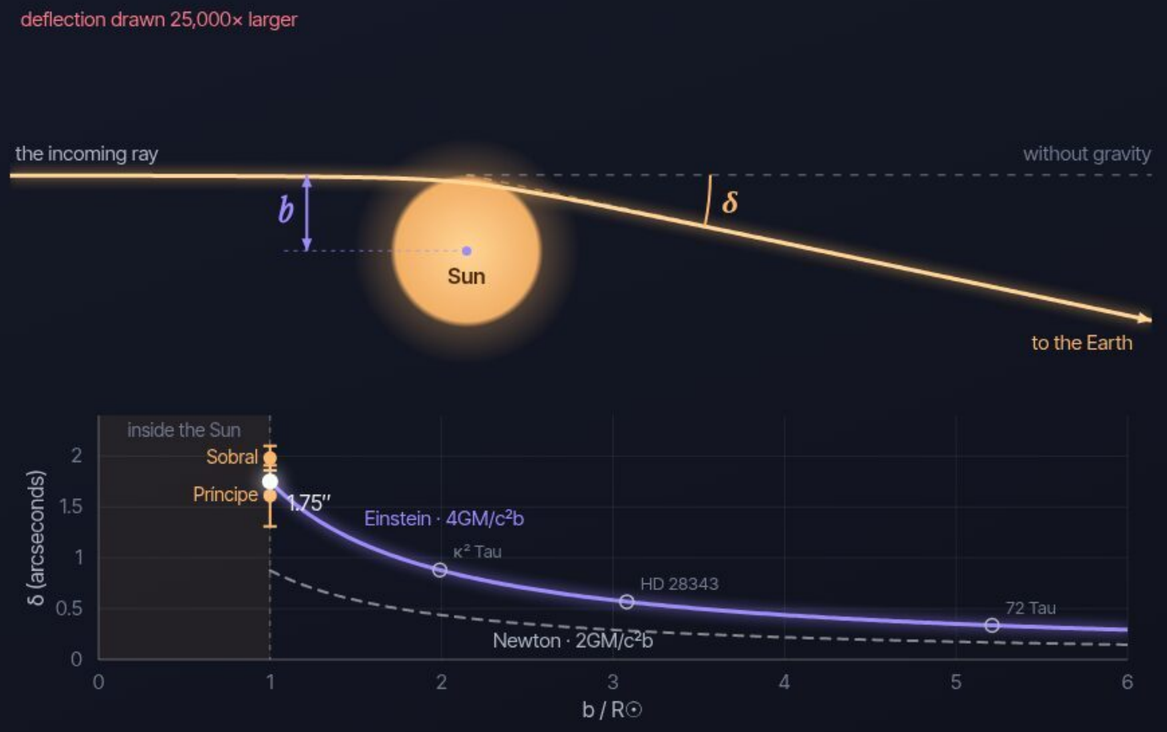
Eclipse: the same field on 29 May, with the Moon covering the Sun. Every star moves away from the Sun, radially.

$$\delta = \frac{4GM}{c^2 b} = 1.75'' \frac{R_\odot}{b}$$

TWICE THE NEWTONIAN VALUE

On the whole plate nothing moves: 1″ is a tenth of a pixel. In the loupes, at true scale, as under the micrometer, the star jumps by a fraction of its own width — and the centre of a blob can be measured far better than its width.

Where *the 1.75″* comes from



$$\delta = \frac{4GM_\odot}{c^2 b} = \frac{2r_s}{b}$$

b: FROM THE CENTRE OF THE SUN TO THE LINE OF THE INCOMING RAY

$$\left[\frac{GM}{c^2 b}\right] = \frac{(\mathrm{m^3\,kg^{-1}\,s^{-2}})(\mathrm{kg})}{(\mathrm{m^2\,s^{-2}})(\mathrm{m})} = 1$$

DIMENSIONLESS: THE ANGLE COMES OUT IN RADIANS

$$b = R_\odot:\ \delta = \frac{5.91\,\mathrm{km}}{6.96\times10^5\,\mathrm{km}} = 8.49\times10^{-6}\,\mathrm{rad}$$

$4GM_\odot/c^2$ = 5.91 km: TWICE THE SUN'S r_s

$$1\,\mathrm{rad} = \frac{180}{\pi}\times 3600'' \approx 206{,}265'' \ \rightarrow\ \delta \approx 1.75''$$

NEWTON, WITH LIGHT AS A PARTICLE: HALF OF IT, 0.87″

1859 – 1915 · THE PERIHELION OF MERCURY

Mercury's orbit *does not close*



PERIHELION ADVANCE	PER CENTURY
observed (relative to the stars)	574″
other planets, by Newton	531″
unexplained remainder	43″

$$\Delta\varphi = \frac{6\pi G M_\odot}{c^2\, a(1-e^2)} = 0.104'' \text{ per orbit}$$

GR · EINSTEIN, NOVEMBER 1915 · 415 ORBITS PER CENTURY → 43.0″

Le Verrier found the remainder in 1859 and proposed a hidden planet, Vulcan. It was never found. Einstein got the 43″ with no adjustable parameter.

> *"For a few days, I was beside myself with joyous excitement."*
>
> Einstein to Ehrenfest, January 1916

DECEMBER 1915 · RUSSIAN FRONT

Schwarzschild and the *first exact solution*



Karl Schwarzschild
1873 – 1916

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \frac{dr^2}{1 - r_s/r} + r^2 d\Omega^2$$

$R_s = 2GM/c^2$ · VALID OUTSIDE ANY NON-ROTATING SPHERICAL BODY

$1\ r_s$	$1.5\ r_s$	$3\ r_s$	2.95 km
event horizon	photon sphere	ISCO: last stable orbit	r_s of the Sun (Earth: 8.9 mm)

He wrote to Einstein on 22 December 1915 while serving as an artillery officer. He died in May 1916.

WHAT WE NEED FROM THE METRIC

Curvature does *three things* to a photon

1 · IT BENDS THE PATH

The photon follows a **null geodesic**. Near the surface of a neutron star the bending is tens of degrees: we see **more than half** of the star.

2 · IT CHANGES THE ENERGY

A photon emitted at r with local energy E reaches infinity with

$$E_\infty = E\sqrt{(1 - R_S/r)}$$

GRAVITATIONAL REDSHIFT

3 · IT DELAYS

Different paths take different times (Shapiro delay): photons from each point of the star arrive tens of microseconds apart.

Ray tracing puts all three into a single calculation, **pixel by pixel**, for each instant of the rotation.

In a neutron star, $R \approx 2.5$–$3\ R_S$: none of the three effects is a small correction.

And all three depend only on the metric outside the star — which is why a light curve carries information about gravity.

Two constants *and a potential*

The metric depends on neither t nor ϕ: energy and angular momentum are conserved.

$$E = \left(1 - \frac{R_s}{r}\right)\dot{t}$$

$$L = r^2\,\dot{\varphi}$$

For light only the ratio matters, the **impact parameter** b = L/E:

$$\frac{1}{b^2} = \frac{\dot{r}^2}{L^2} + V(r), \quad V = \frac{1}{r^2}\left(1 - \frac{R_s}{r}\right)$$

A "POTENTIAL" FROM GEOMETRY ALONE

$$\left(\frac{du}{d\varphi}\right)^2 = \frac{1}{b^2} - u^2(1 - R_s u)$$

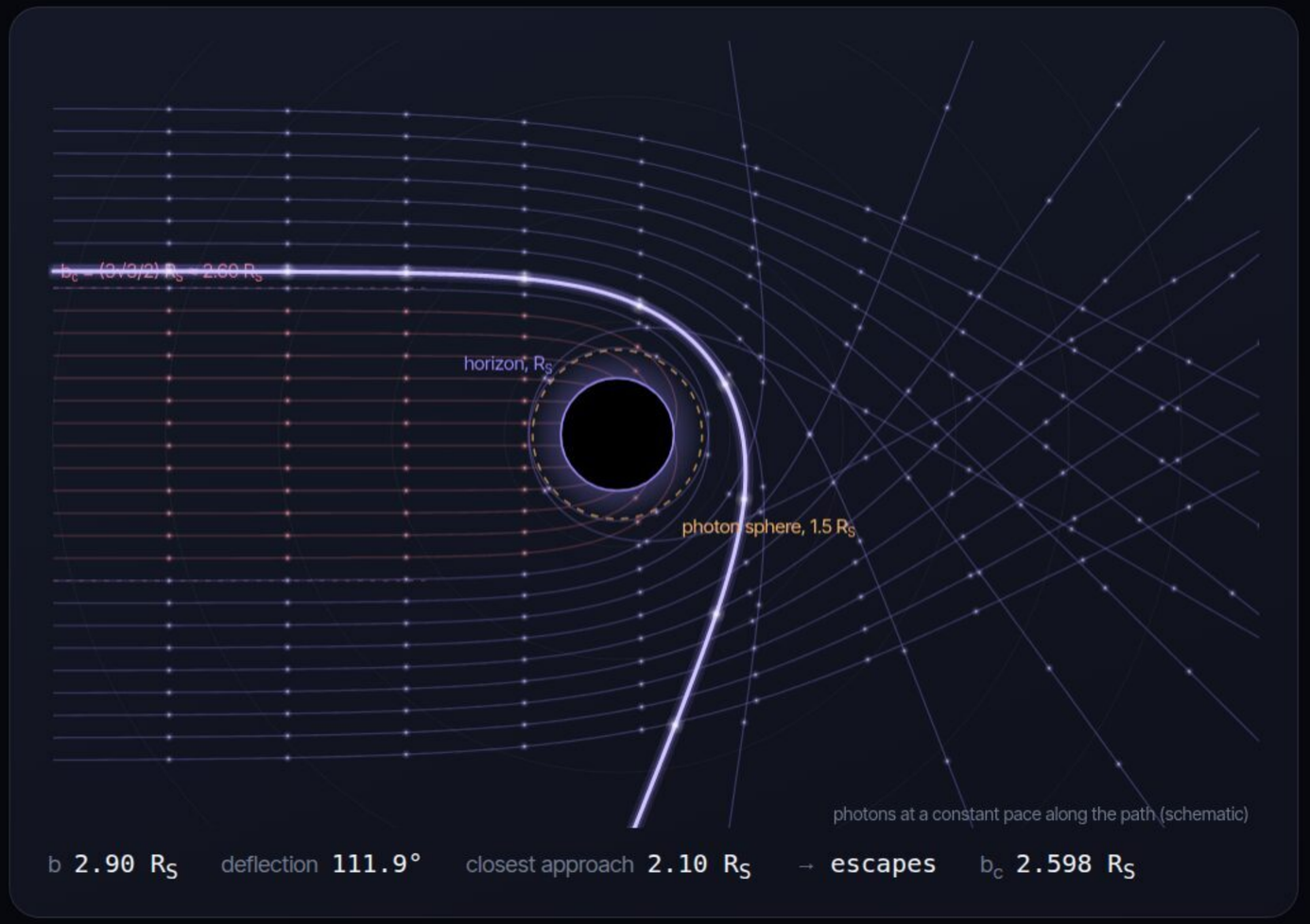
THE ORBIT · $U = 1/R$

The maximum of V lies at $r = 1.5\,R_S$: the **photon sphere**. Below $b_c = (3\sqrt{3}/2)\,R_S$ the ray comes from the horizon — or from the star.



SCHWARZSCHILD GEODESICS

How light *bends*



$$u'' + u = \frac{3}{2} R_S u^2$$

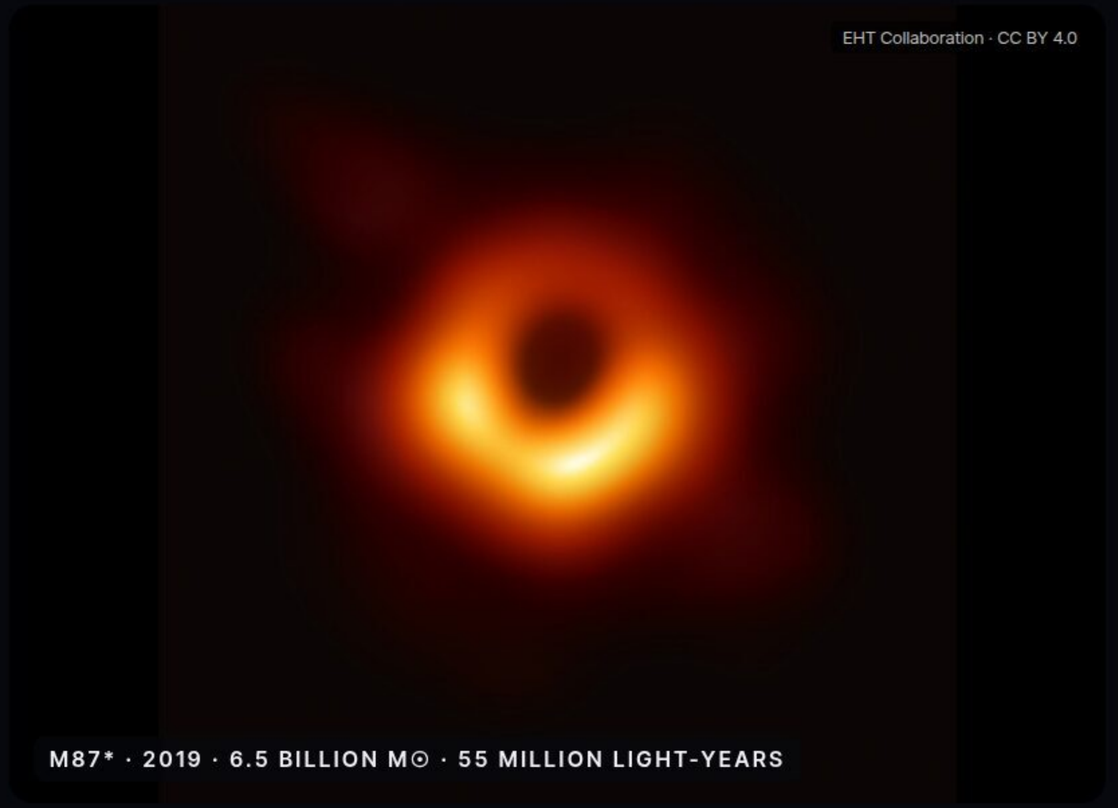
$U = 1/R$ · ORBIT EQUATION FOR LIGHT

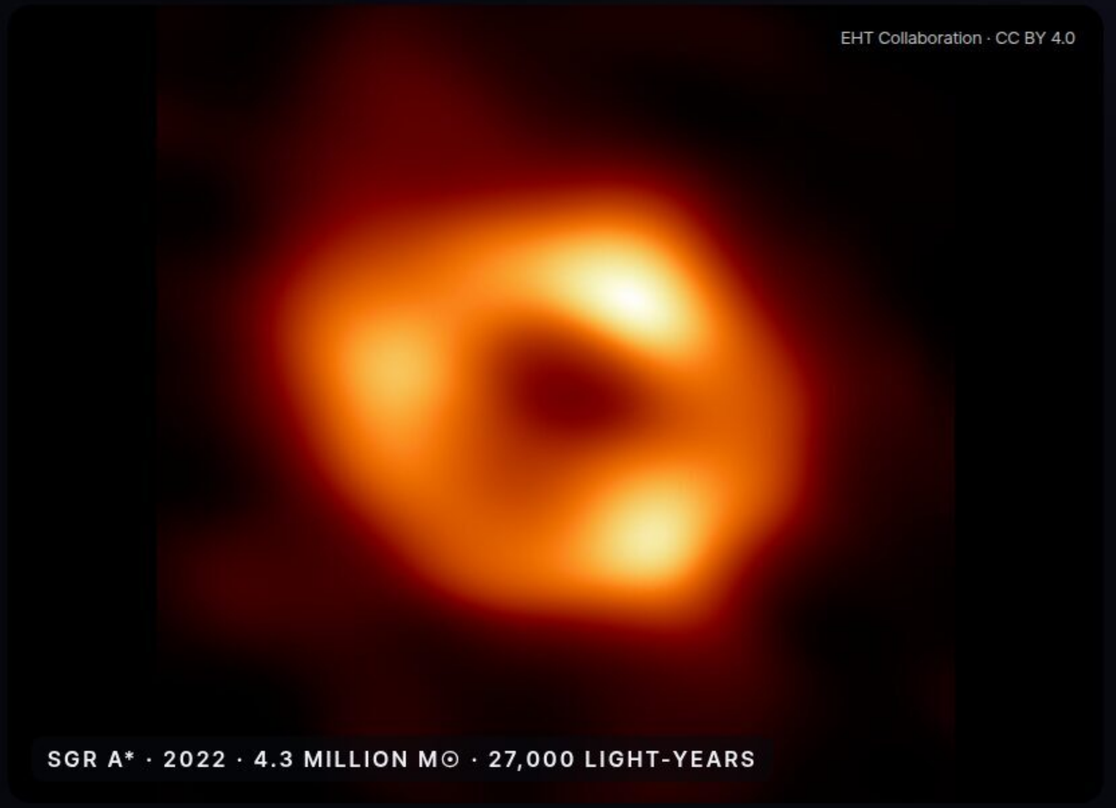
- $b > 3\sqrt{3}\,GM/c^2 \approx 2.6\,R_S$: the light is deflected and escapes.
- $b = 2.6\,R_S$: unstable orbit on the **photon sphere**, $r = 1.5\,R_S$.
- $b < 2.6\,R_S$: captured.

The term on the right is the General Relativity correction. Without it, light would travel in a straight line.

EVENT HORIZON TELESCOPE · 2019 AND 2022

The *shadows*





The ring is ~5 R_S across: it is the photon sphere, at 1.5 R_S, seen through the lens of the black hole itself — shadow diameter = $2\sqrt{27}\,GM/c^2 \approx 5.2\,R_S$.

COMPACTNESS

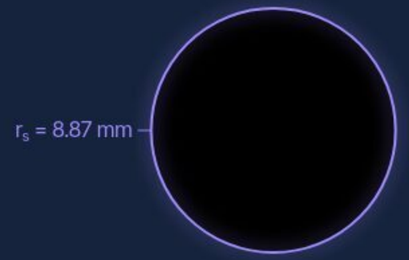
Where GR *stops being a correction*

THE RADIUS AND THE SCHWARZSCHILD RADIUS OF THE SAME MASS

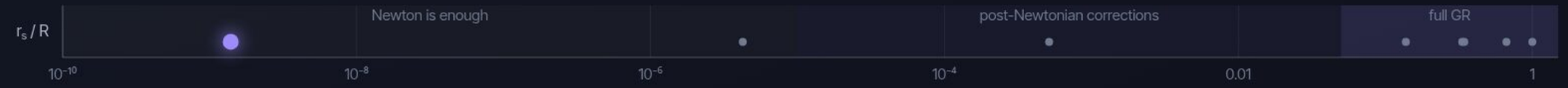
Earth | Sun | Sirius B | proto-NS | neutron star | ISCO | photon sph. | horizon

Earth — R = 6,371 km · the edge is off the frame

squeezed down to r_s, the Earth would be a black hole the size of a marble



mass · radius	5.97×10^{24} kg · 6,371 km
Schwarzschild radius, $r_s = 2GM/c^2$	8.87 mm
compactness, r_s/R	1.39×10^{-9}
escape velocity	11.2 km/s
surface gravity	1 g
redshift, z	6.96×10^{-10}
the surface clock loses	60.1 µs per day
grazing-ray deflection	5.7×10^{-4}″
surface seen from afar	50% + 7.0×10^{-8}%



Surface compared with infinity: $z = (1 - r_s/R)^{-1/2} - 1$, $g = (GM/R^2)(1 - r_s/R)^{-1/2}$ · deflection from the exact orbit integral (weak field: $2r_s/R$) · ISCO, photon sphere and horizon: a 1.4 M☉ star of that size · the objects advance on their own until the first click.

THE ANGLE THAT MATTERS

From the surface to infinity: $\psi(\alpha)$

A photon leaves the surface at an angle α to the normal. By the time it reaches the observer, it has turned through ψ around the centre:

$$b = \frac{R \sin \alpha}{\sqrt{(1 - R_s/R)}}$$

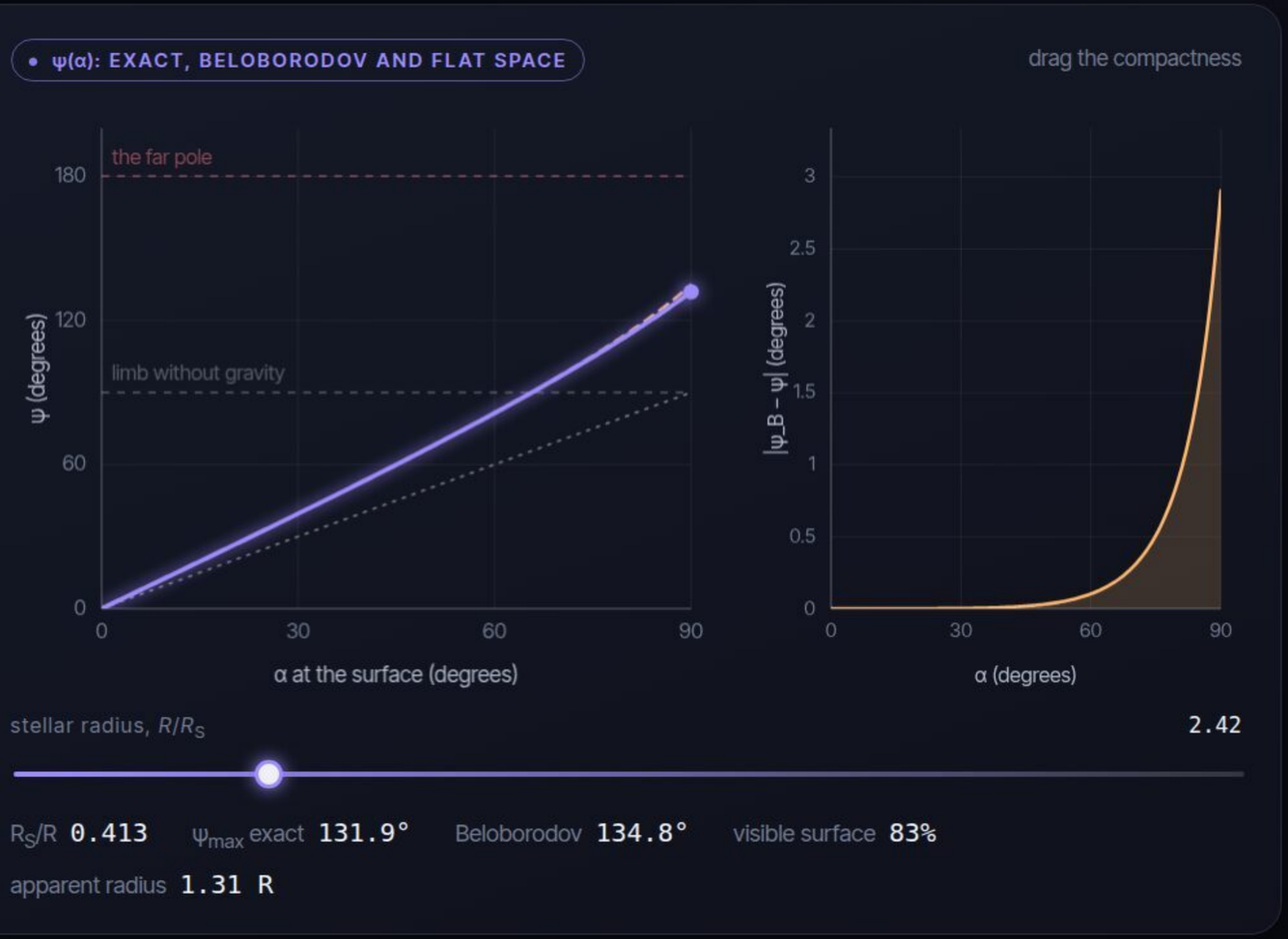
$$\psi(b) = \int_0^{1/R} \frac{du}{\sqrt{(b^{-2} - u^2(1 - R_s u))}}$$

EXACT

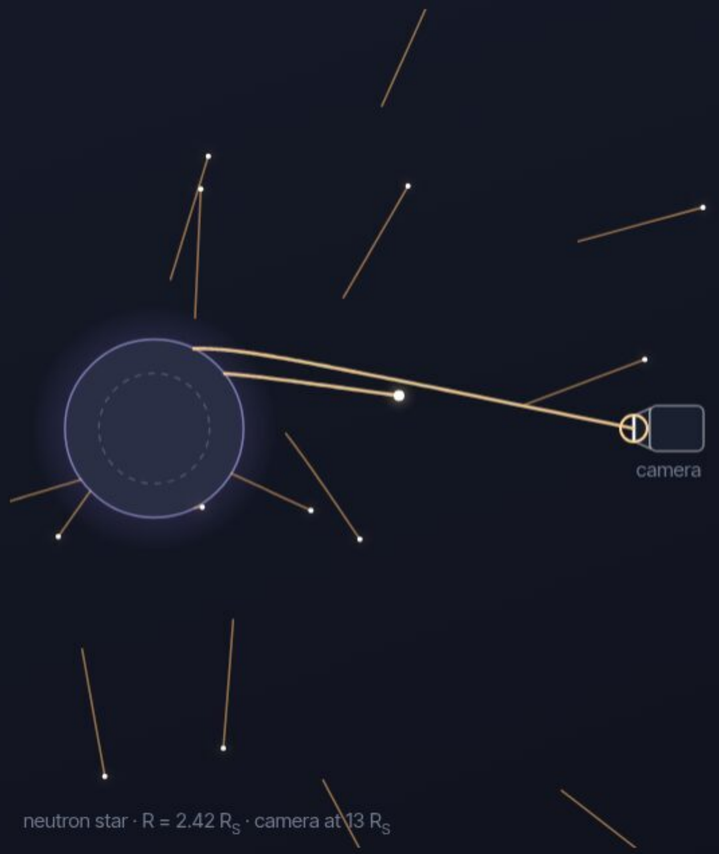
$$1 - \cos \alpha \approx (1 - \cos \psi)(1 - \frac{R_s}{R})$$

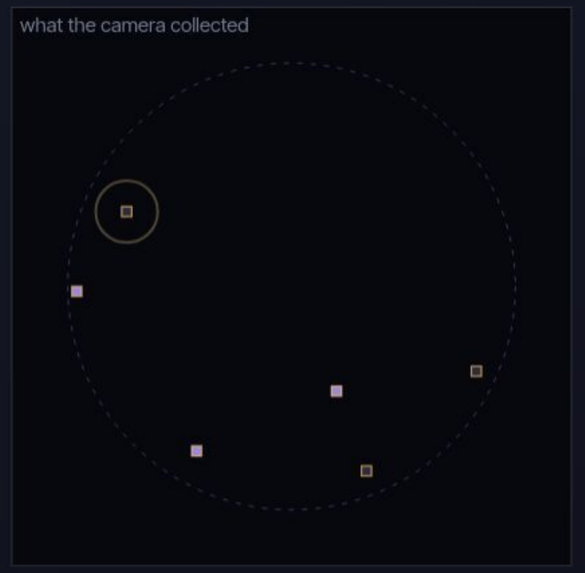
BELOBORODOV (2002)

Without gravity, ψ = α and the limb is at 90°. With gravity, the limb moves further round: **we see behind the star**. The approximation is excellent down to $R \approx 3\ R_S$ and fails precisely where the secondary image appears.



Many rays, *hardly any pixels*



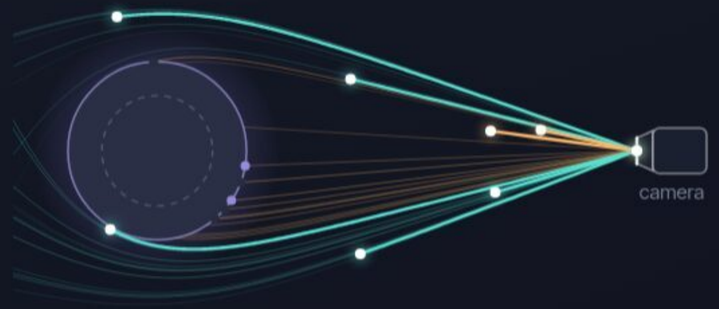


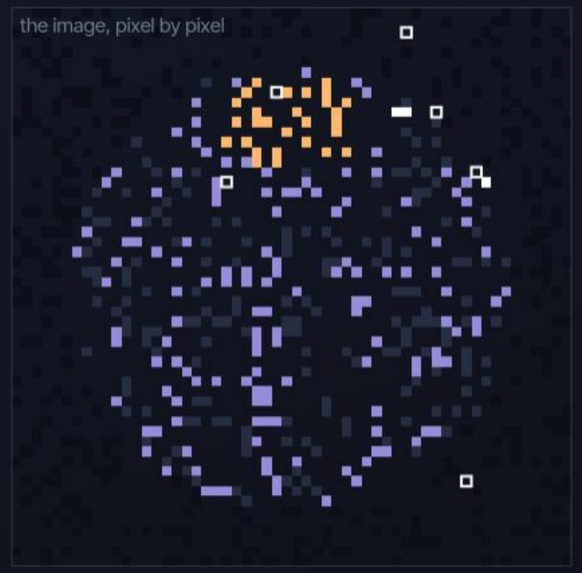
rays launched **33,333** entered the camera **7** (0.021%) pixels lit **6** of 1,568 full image in **~5 h**

- Every point of the surface emits in **every direction**; each ray is followed along its geodesic, outwards.
- Only those that pass through the camera aperture become pixels: here, **2 in every 10,000**.
- Completing this image would take ~10^8 rays: hours of computing.
- With a real telescope, 1 m^2 at 1 kpc, **1 photon in every 10^{40}** arrives.

3D count: uniform point on the surface, isotropic outward direction, exact geodesic to the camera, at 13 R_S. Only a sample is drawn, in the plane; the rays that enter the camera are all shown.

One pixel, *one ray*





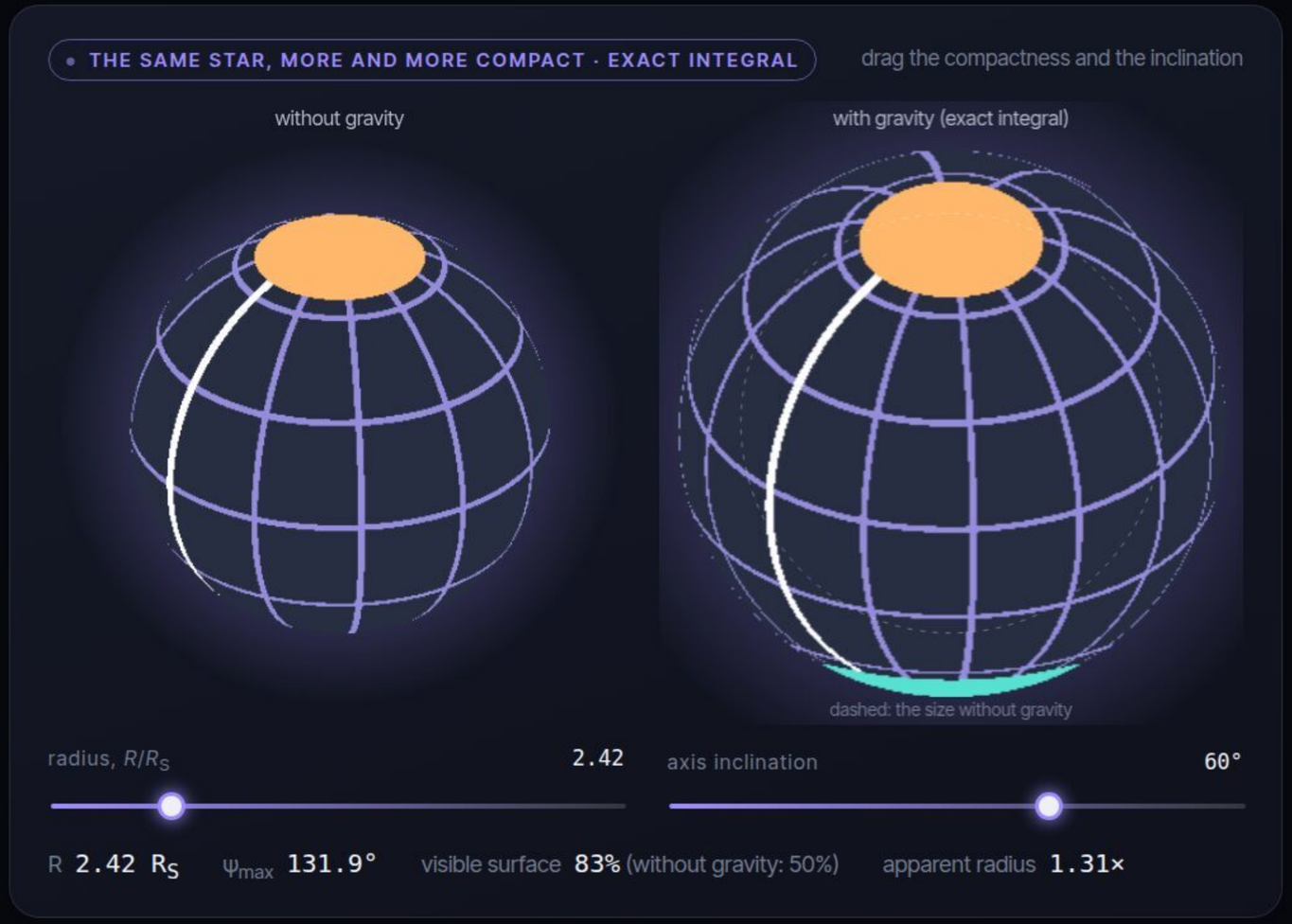
neutron star · R = 2.42 R_S · camera at 13 R_S

pixel (39, 2) · θ 16.9° · b 3.92 R_S escapes: sky rays computed = pixels 684 (22%)

- The rays leave the **camera**, one per pixel: the path of light travelled backwards. The direction of the pixel gives the impact parameter *b*.
- The geodesic is integrated until the ray escapes to the sky or hits the surface at some point, at an angle α.
- The pixel takes the colour of the point the light left from — corrected for redshift and for the emission angle.
- Every ray computed becomes a pixel: **3,136 rays, 3,136 pixels**, against the 10^8 of the previous slide.

With spherical symmetry, ψ depends only on *b*: one table ψ(*b*), computed once, serves every pixel. Each ray is rotated into the plane of the drawing.

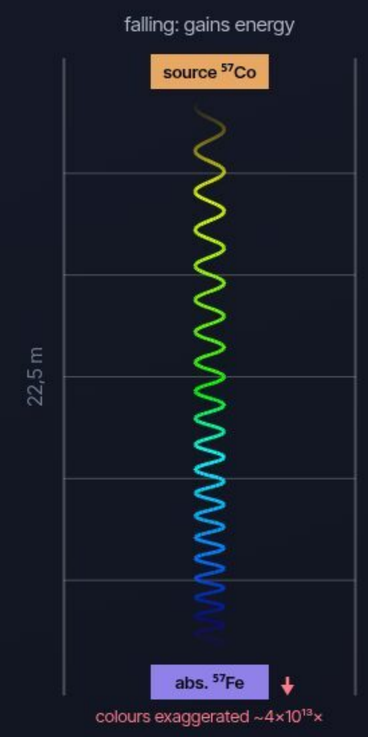
We see *behind* the star

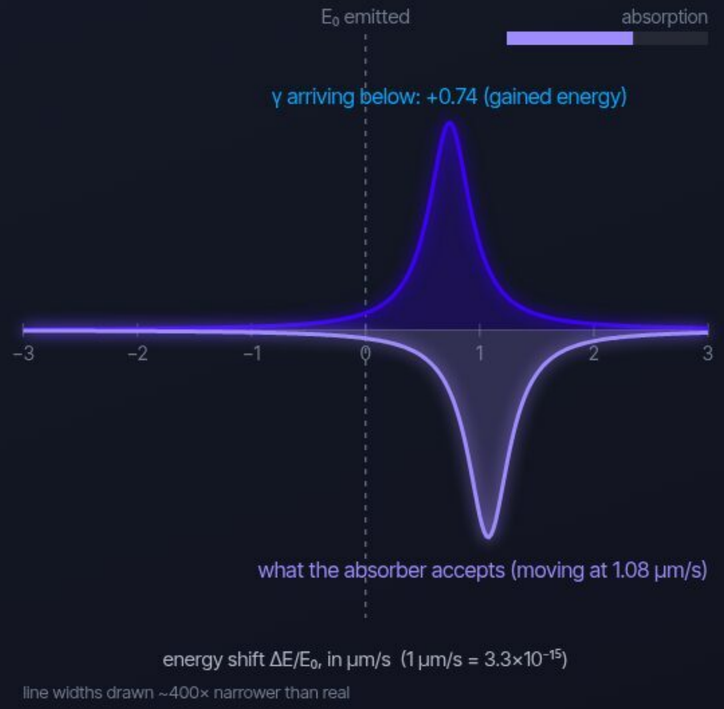


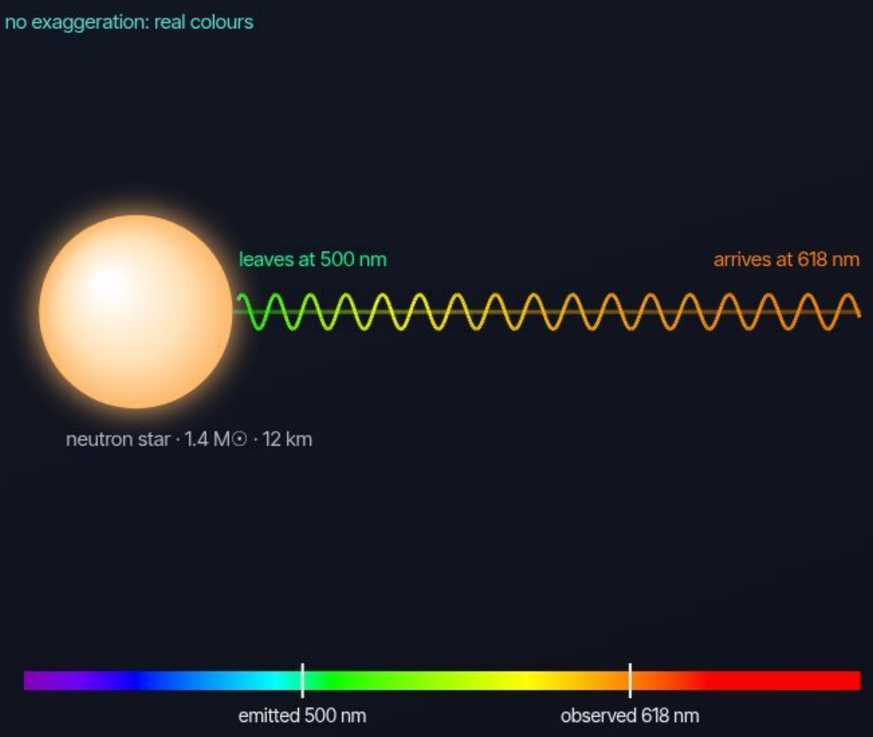
On the left, the star as it would look without gravity; on the right, ray-traced. The grid is the same; gravity bends the light around the star and reveals the **far hemisphere**.

- The image is larger: apparent radius $R/\sqrt{(1 - R_S/R)}$.
- For $R < 1.76\ R_S$ the whole surface is visible — the far pole becomes a ring.
- Below $1.5\ R_S$ the star would lie inside the photon sphere.

Light *loses energy as it climbs*







tower $\Delta\nu/\nu = gh/c^2$ **2.46×10⁻¹⁵** resonance at **+0,74 μm/s** ≈ 2.7 mm per hour absorber now **1.08 μm/s** star $1 + z = (1 - r_s/R)^{-½}$ **1.235**

14.4 keV γ rays from ⁵⁷Fe (Mössbauer effect). Pound & Rebka, PRL 4, 337 (1960): ~10%; Pound & Snider, Phys. Rev. 140, B788 (1965): ~1%. Every 12 s source and absorber swap places, as in the experiment.

The spectrum that *climbs out of the well*

Along a ray, I_ν/ν^3 is conserved. With $g = \sqrt{(1 - R_S/R)}$:

$$I_{obs}(E) = g^3 I_{em}(E/g)$$

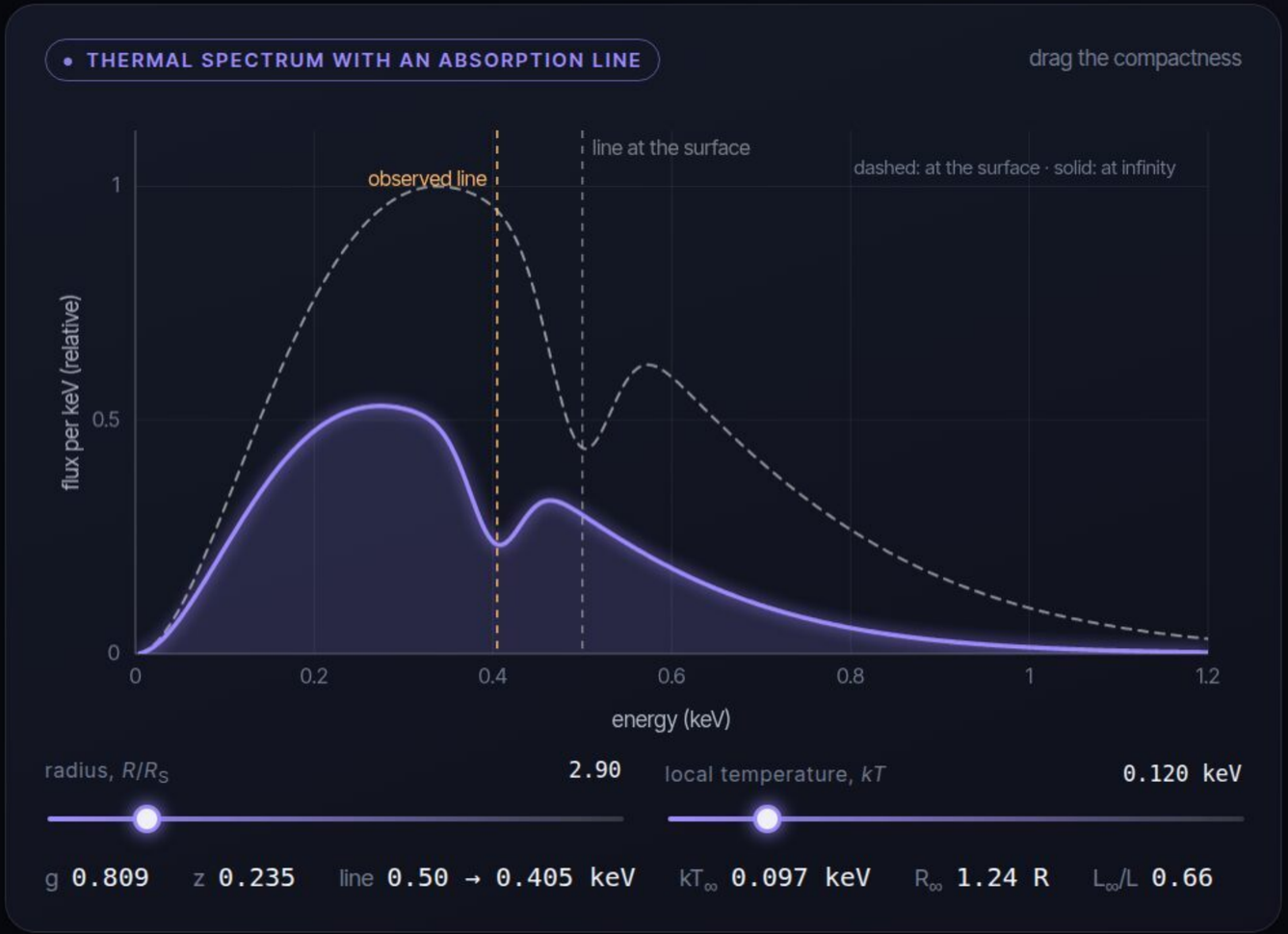
SPECIFIC INTENSITY

$T_\infty = g\,T$

$R_\infty = R/g$

A black body stays a black body, only cooler. The luminosity seen from afar is $L_\infty = g^2 L$.

A **spectral line** of known energy E_0 arrives at $g\,E_0$: measuring the line means measuring the **compactness**.



PART I IN SUMMARY

The *ray tracing* recipe

1. **Choose the exterior metric**: Schwarzschild; Hartle–Thorne or Kerr with rotation; a different one for a different theory of gravity.
2. **Build the table** $\psi(b)$ and the delay $\Delta t(b)$ by integrating the null geodesics once.
3. **For each pixel** (or each surface element): $b \to$ emitting point, angle α, redshift g, delay.
4. **Intensity**: $I_{obs} = g^3\, I(E/g, \alpha')$, with the atmosphere's beaming and the Doppler shift from rotation.
5. **Add up**: image, flux — and, phase by phase, the **light curve** and the phase-resolved spectrum.

EXACT OR APPROXIMATE?

- **Exact integral** (or elliptic integrals): always valid, secondary images included.
- **Beloborodov (2002)** and Poutanen & Beloborodov (2006): closed formulae for ψ and for the delay, excellent for $R \gtrsim 3\ R_S$.
- **Oblate Schwarzschild** (Morsink et al. 2007): a star flattened by rotation, for millisecond pulsars.

This is the engine behind NICER's mass and radius measurements — and behind the results of Part III.

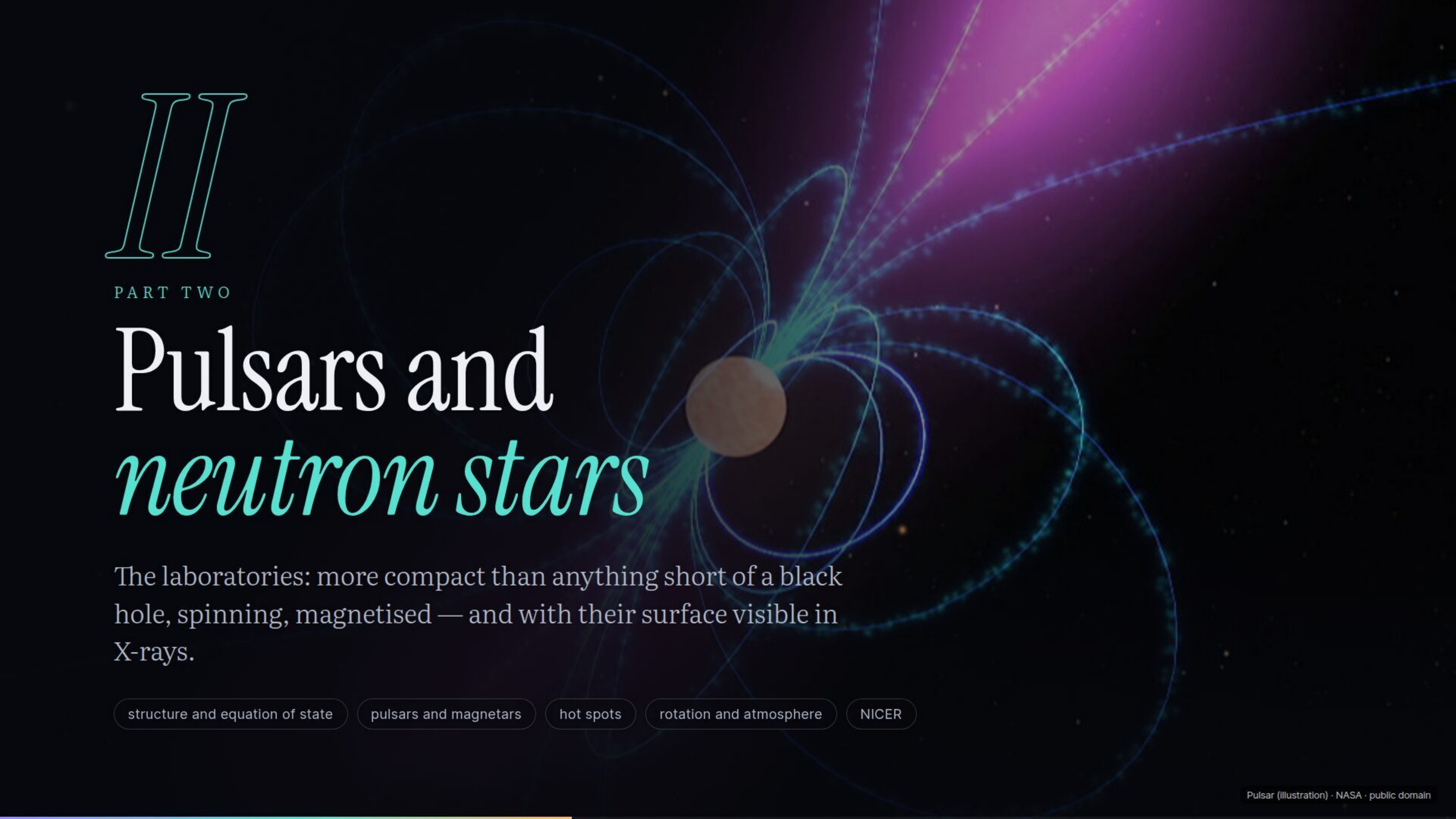
II

PART TWO

Pulsars and *neutron stars*

The laboratories: more compact than anything short of a black hole, spinning, magnetised — and with their surface visible in X-rays.

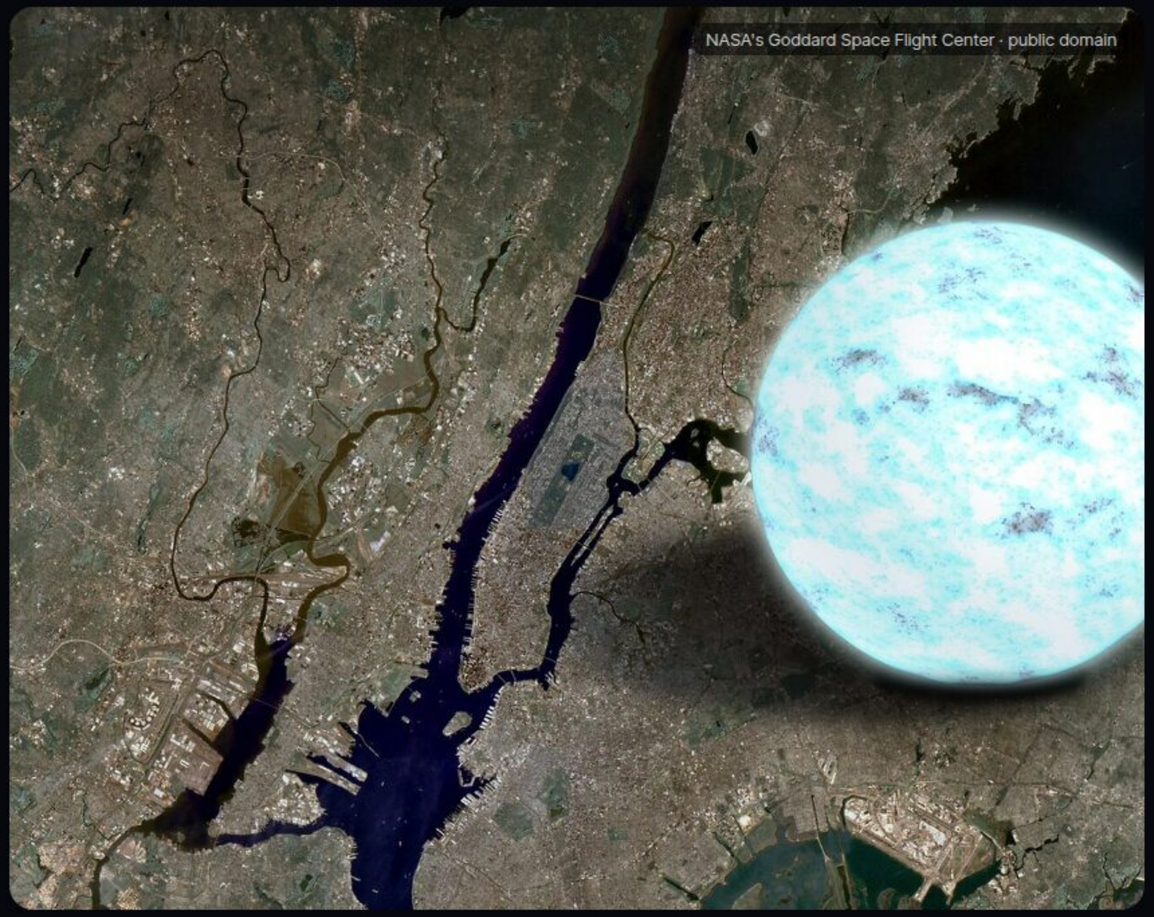
structure and equation of state | pulsars and magnetars | hot spots | rotation and atmosphere | NICER



Pulsar (illustration) · NASA · public domain

NEUTRON STARS IN NUMBERS

A solar mass *in a city*



A neutron star ~20 km across over Manhattan, to scale.

1.1–2.1 $M_\odot$
measured masses

11–13 km
radius (NICER, gravitational waves)

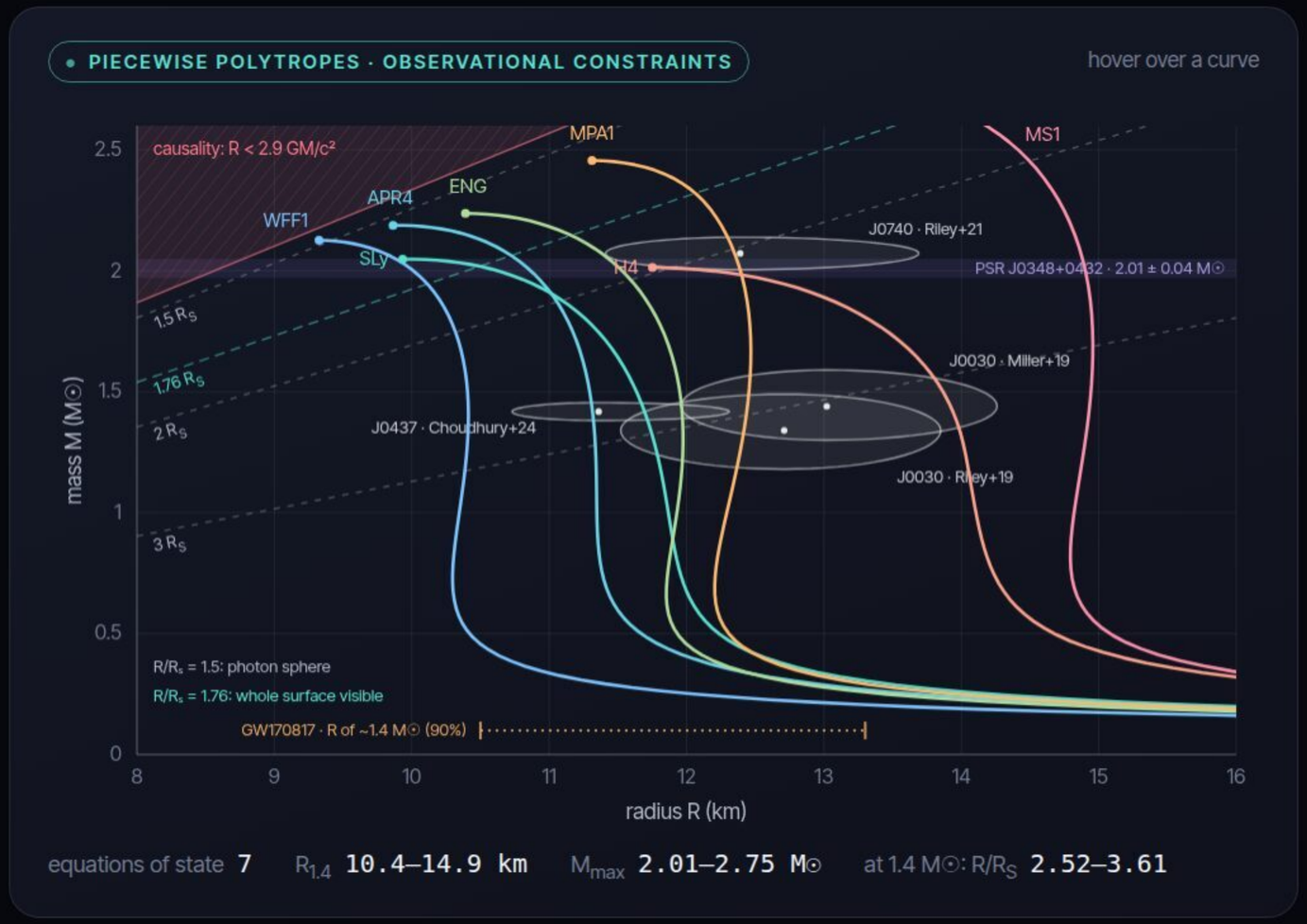
2–3 R_S
compactness: the regime of Part I

10^8–10^{15} G
surface magnetic field

Central density a few times that of the atomic nucleus, ~10^{15} g/cm^3. Spin periods from **1.4 ms** to tens of seconds.

Surface gravity is ~10^{11} times the Earth's; the escape velocity, 0.5 to 0.7 **c**.

From nuclear physics *to the M–R curve*



$$\frac{dP}{dr} = -\frac{G(\varepsilon + P)(m + 4\pi r^3 P/c^2)}{c^2\, r(r - 2Gm/c^2)}$$

RELATIVISTIC HYDROSTATIC EQUILIBRIUM (TOV)

Each equation of state P(ε) gives a mass–radius curve and a maximum mass. Measuring the **mass and radius of the same star** picks between them.

The lines of constant compactness link the curve to Part I: at $R = 1.76\ R_S$ the whole surface would be visible; at $1.5\ R_S$, the photon sphere.

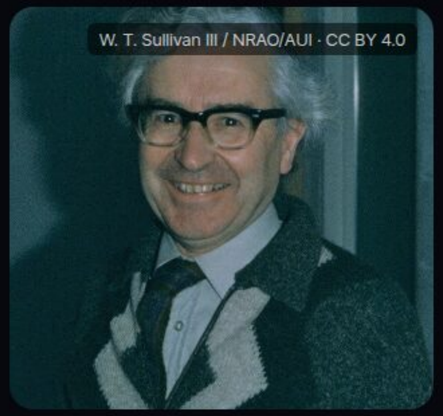
Piecewise polytropes of Read, Lackey, Owen & Friedman (2009); constraints (68%): NICER — Riley et al. 2019, 2021; Miller et al. 2019; Choudhury et al. 2024; PSR J0348+0432 (Antoniadis et al. 2013); GW170817 (Abbott et al. 2018).

CAMBRIDGE · 28 NOVEMBER 1967

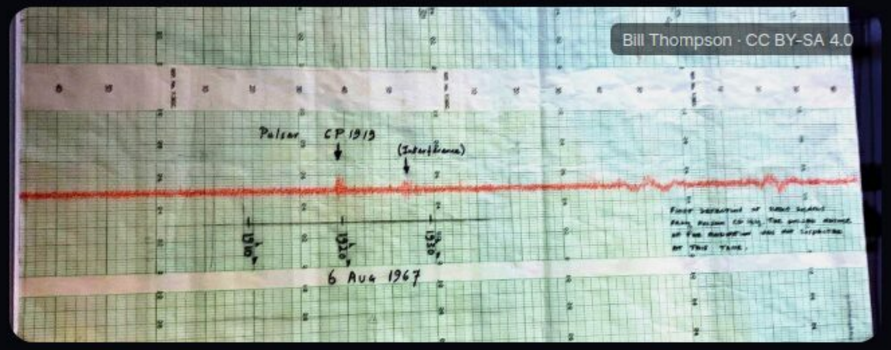
The first pulsar: *"LGM-1"*



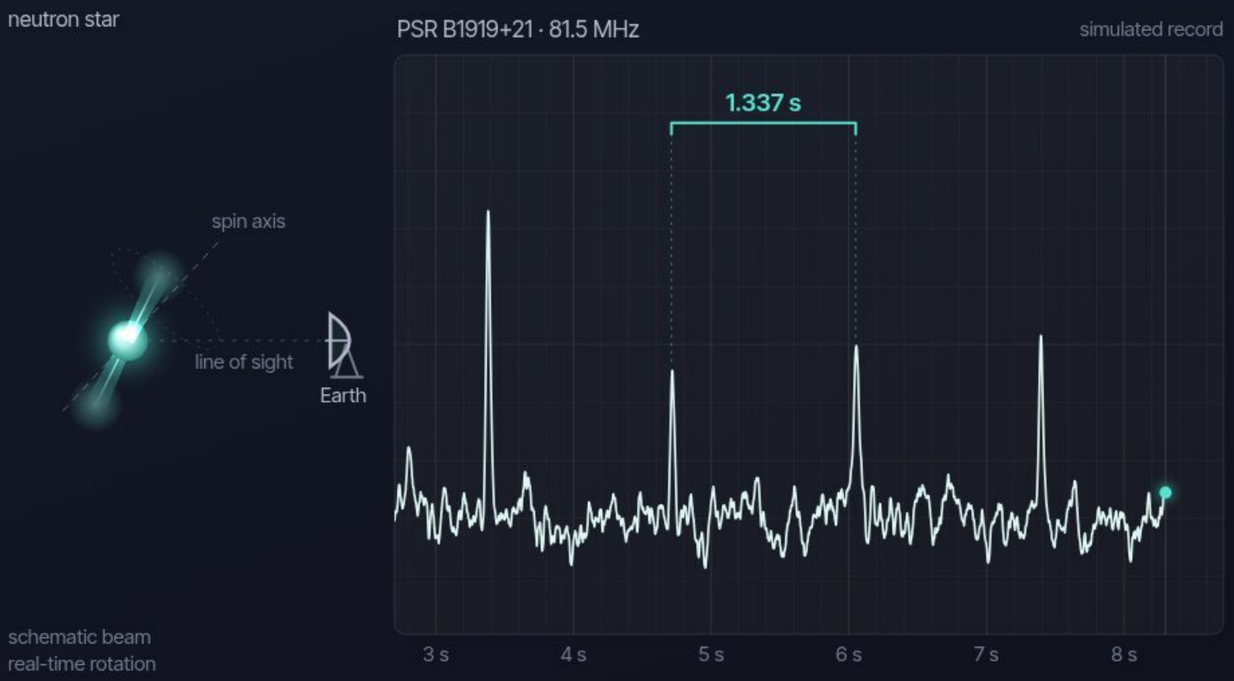
Jocelyn Bell
in 1967, PhD student



Antony Hewish
supervisor · Nobel Prize 1974



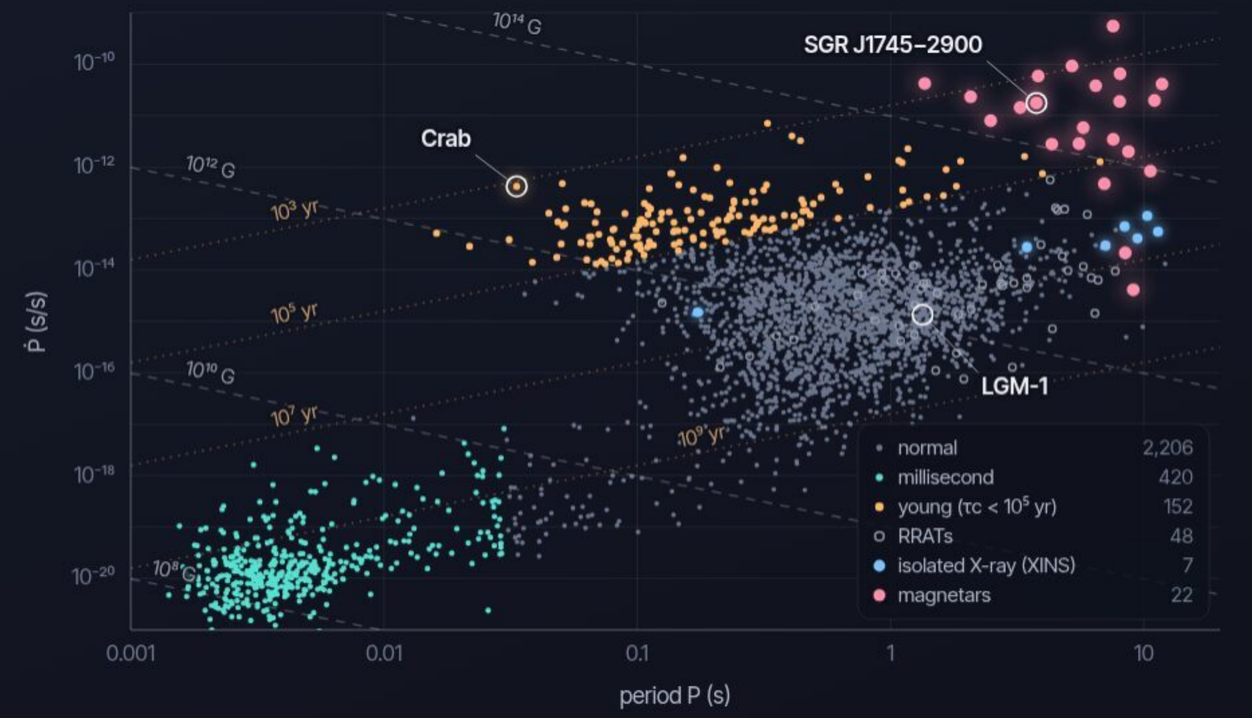
The original chart, August 1967: the "scruff" pencilled in as CP 1919.



period **1.337 302 s** Ṗ **1.35×10⁻¹⁵ s/s** frequency **81.5 MHz** pulses recorded **6**

Bell noticed a bit of "scruff" that kept coming back on the radio telescope's paper chart: pulses every **1.337 s**, as regular as clockwork. Nickname: **LGM-1**, for *Little Green Men*. It was a spinning neutron star.

What is measured, *and what is inferred*



OBSERVABLE	EXPRESSION
period	P
characteristic age	$t_c = P/2\dot{P}$
magnetic field	$B = (3Ic^3P\dot{P}/8\pi^2R^6)^{1/2}$
rotational energy lost	$\dot{E}_{rot} = -4\pi^2I\dot{P}/P^3$

CANONICAL NEUTRON STAR

$M = 1.4\ M_\odot \cdot R = 10\ \text{km} \cdot I = 10^{45}\ \text{g cm}^2$

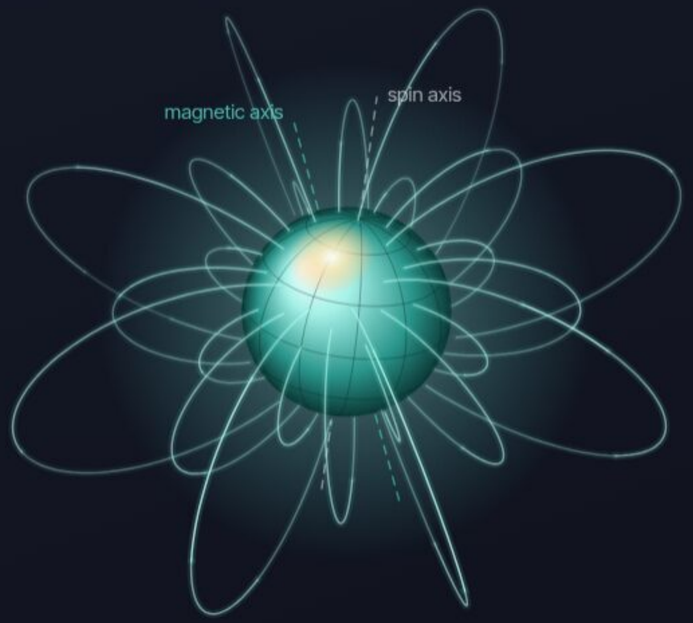
$B = 3.2\times10^{19}\ (P\dot{P})^{1/2}$ G

$|\dot{E}_{rot}| = 3.9\times10^{46}\ \dot{P}/P^3$ erg/s

Only P and $\dot{P}$ are measured. The rest comes from assuming a spinning magnetic dipole — the straight lines in the diagram.

A magnetar, *spinning*

SGR J1745−2900



field lines of a tilted dipole · real-time rotation

illustrative bursts

P 3.76 s B ≈ 2.6×10¹⁴ G characteristic age ≈ 3,400 yr $\dot{E}_{rot}$ ≈ 1.3×10³⁴ erg/s

bursts seen 1

EVENT	ENERGY
short bursts (~0.1 s)	10^{39} – 10^{41} erg
outbursts (months)	10^{41} – 10^{43} erg
giant flares	10^{44} – 10^{46} erg

The field lines, anchored in the crust, get twisted over time. When the crust can no longer take the stress, it cracks — a **starquake** — and the magnetic energy comes out all at once.

Giant flares in the Galaxy and the Magellanic Clouds: 1979, 1998 and 2004.

Why *hot spots* appear

MILLISECOND PULSARS

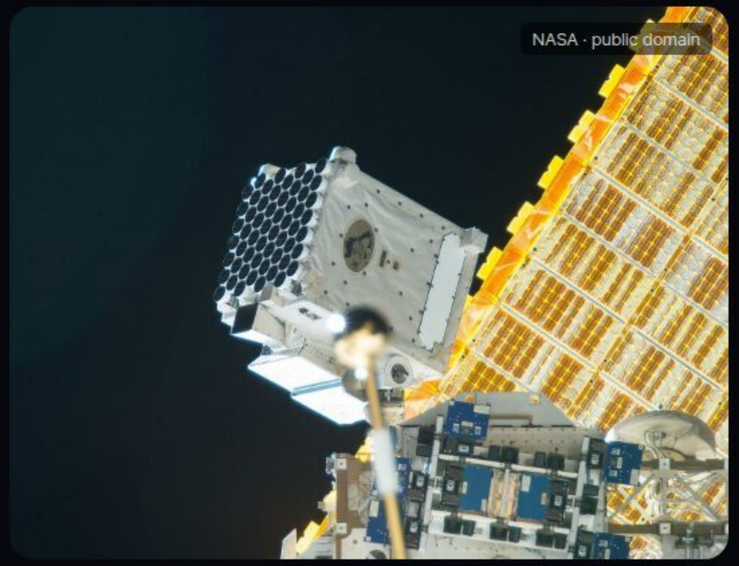
Particles accelerated in the magnetosphere flow back and **heat the polar caps** to ~10^6 K. They are NICER's targets.

MAGNETARS

Field decay and the currents along twisted field lines heat regions of the crust — which change after each *outburst*.

ISOLATED COOLING STARS

The magnetic field makes **heat conduction anisotropic**: poles hotter than the equator (the "Magnificent Seven").



NICER · on the Space Station since 2017 · ~100 ns time resolution



XMM-Newton · ESA, since 1999 · the largest area in soft X-rays



Chandra · NASA, since 1999 · the sharpest angular resolution

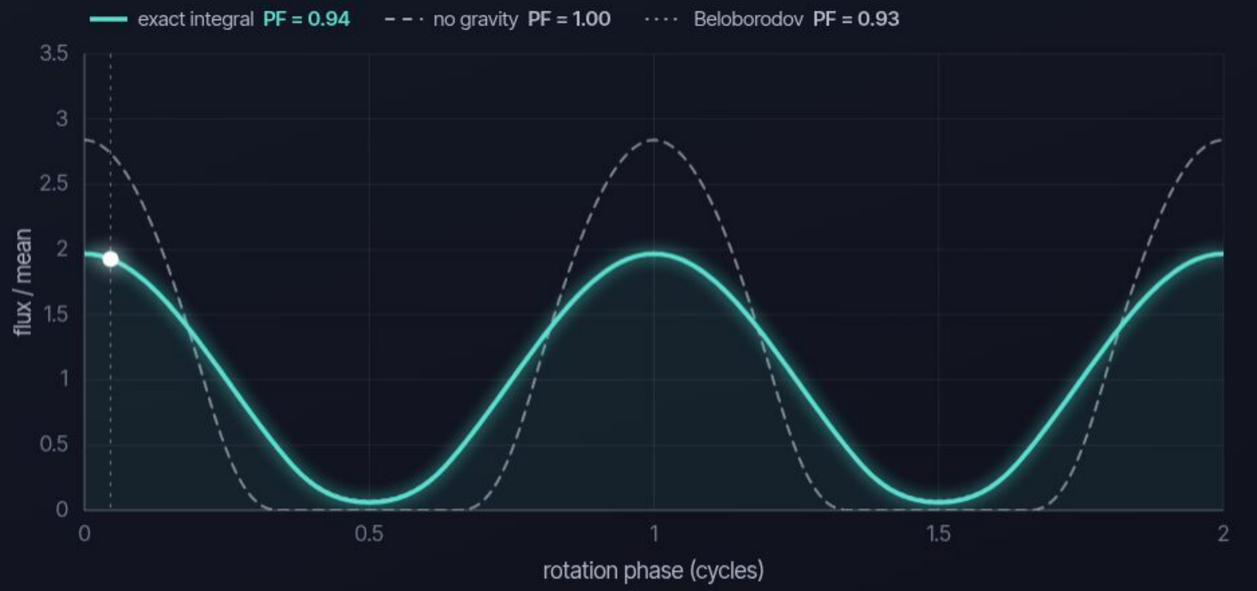
One spot, *one light curve*

• STATIC STAR · EXACT INTEGRAL, BELOBORODOV OR FLAT SPACE

no gravity | Beloborodov | exact integral



static star · 1 turn per 4 s
sizes in units of R



radius, R/R_S 2.42 | spot ↔ axis, i 60° | observer ↔ axis, j 75° | angular radius of the spot 20°

PF no gravity 1.000 PF Beloborodov 0.928 PF exact 0.939 surface visible 83% ψ_{max} 131.9° class B · centre hidden 10% of the turn (the rim stays)

secondary image no (needs $R < 1.76\ R_S$)

Antipodal, *or not*

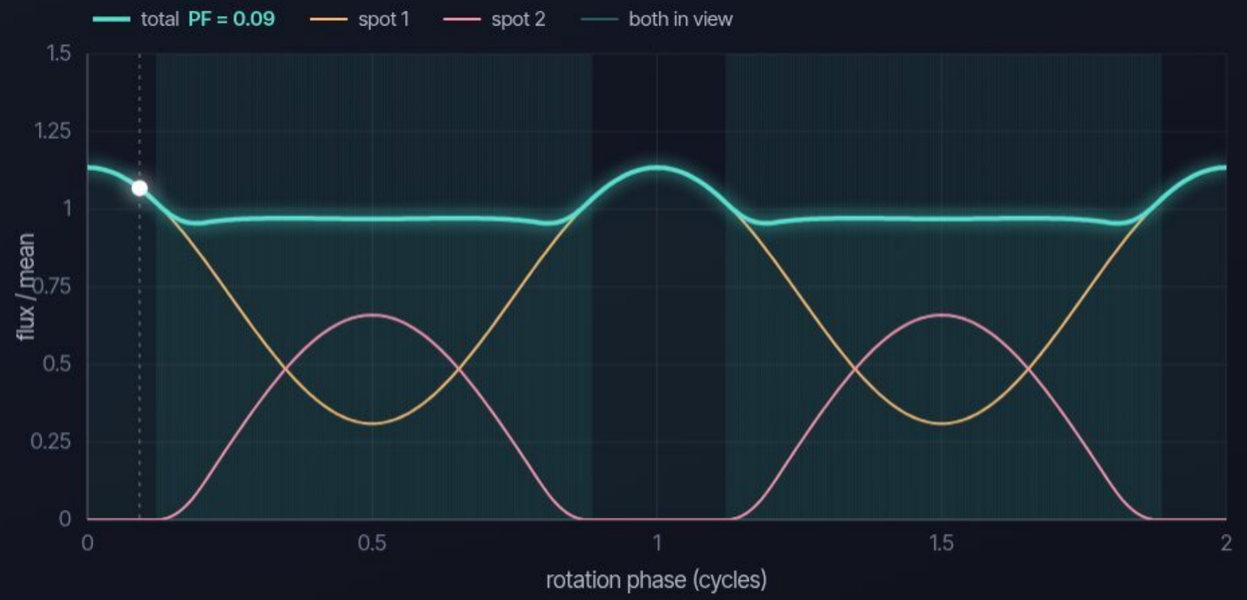
• TWO SPOTS: ONE AT EACH MAGNETIC POLE

antipodal | offset

• spot 1
• spot 2



1.4 M⊙ · 10 km · R = 2.42 R_s
exact integral · 1 turn per 4 s



magnetic axis ↔ rotation, *i* 45°

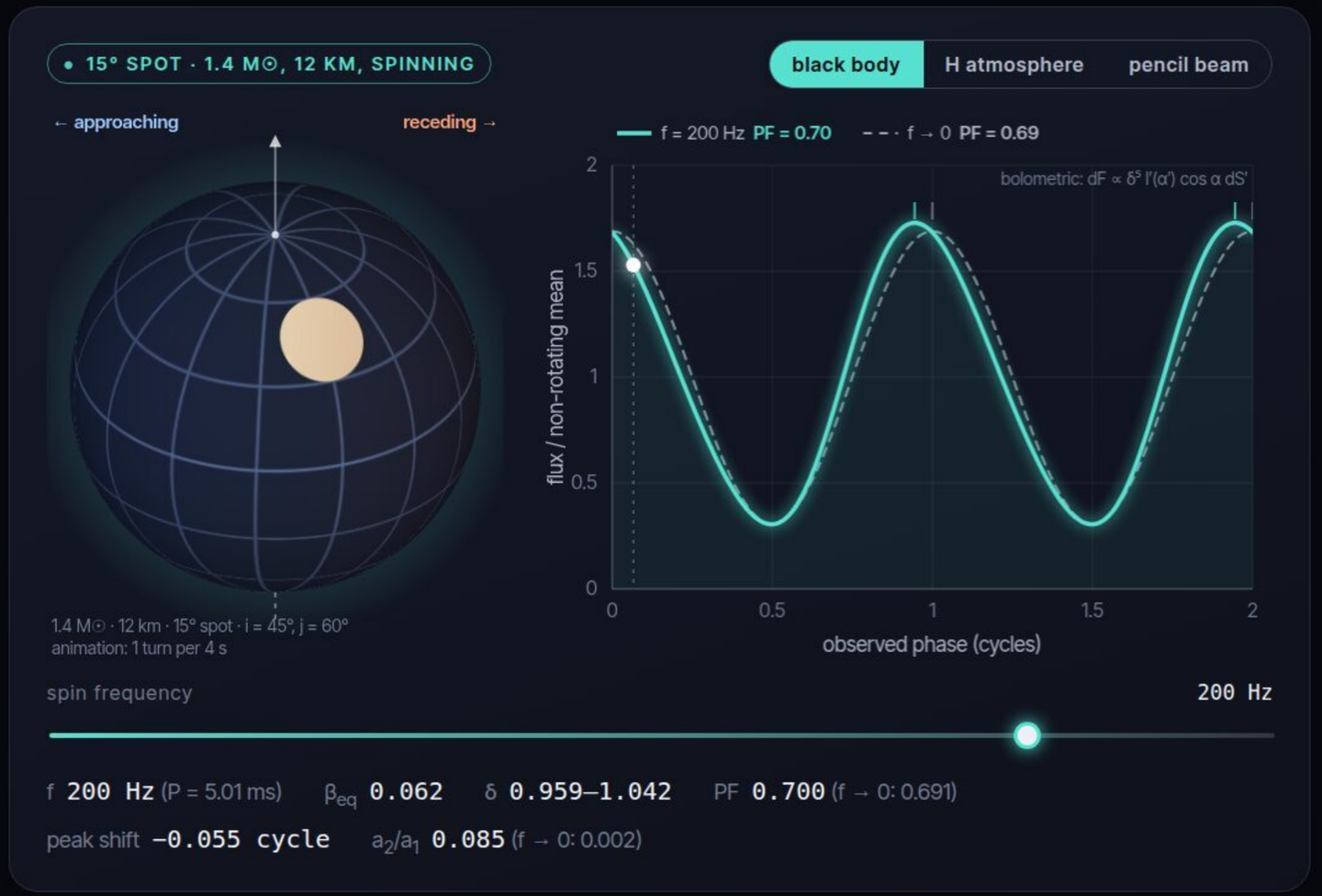
observer ↔ rotation, *j* 60°

offset of the second spot 0°

class **II** · primary always; antipodal sometimes PF **0.086** spot 1 in view **always** spot 2 in view **77% of the turn** both **77%** pulses separated by **0.50 cycle**

Ψ_{max} **131.9°**

Doppler, aberration, delays *and beaming*

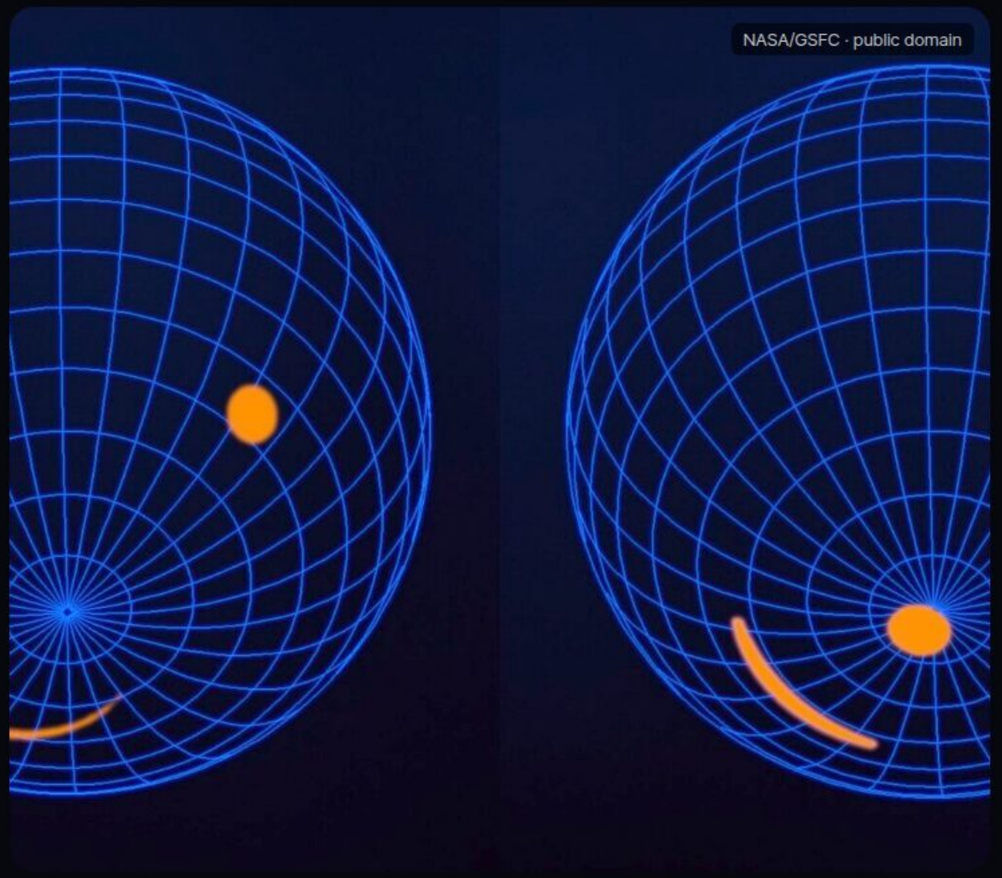


- **Doppler**: the surface moves at up to 0.1–0.2 c; the approaching side is brighter (bolometric flux ∝ δ^5).
- **Aberration**: the emission angle changes in the frame of the surface.
- **Delays**: tens of μs between points on the star — fractions of the period at hundreds of Hz.
- **Beaming**: an atmosphere does not emit equally in all directions; the profile depends on it.

Together they make the pulse **asymmetric** and create harmonics — and the asymmetry measures R separately from M.

NICER · PULSE-PROFILE MODELLING

Mass and radius *from the pulse shape*



PSR J0030+0451 · the solutions of the two teams: Amsterdam (Riley+ 2019, left) and Maryland (Miller+ 2019). In both, the spots are far from antipodal.

PULSAR	MASS ($M_\odot$)	RADIUS (KM)
PSR J0030+0451 · Riley+ 2019	$1.34^{+0.15}_{-0.16}$	$12.7^{+1.1}_{-1.2}$
PSR J0030+0451 · Miller+ 2019	$1.44^{+0.15}_{-0.14}$	$13.0^{+1.2}_{-1.1}$
PSR J0740+6620 · Riley+ 2021	$2.07^{+0.07}_{-0.07}$	$12.4^{+1.3}_{-1.0}$
PSR J0437−4715 · Choudhury+ 2024	1.42 ± 0.04	$11.4^{+1.0}_{-0.6}$

NICER time-stamps each photon to ~100 ns. Folding millions of photons at the period gives the pulse profile; ray tracing, with M, R, the geometry and the spots left free, fits that profile by Bayesian inference.

J0030's spots came out non-antipodal and elongated: the field is not a centred dipole.

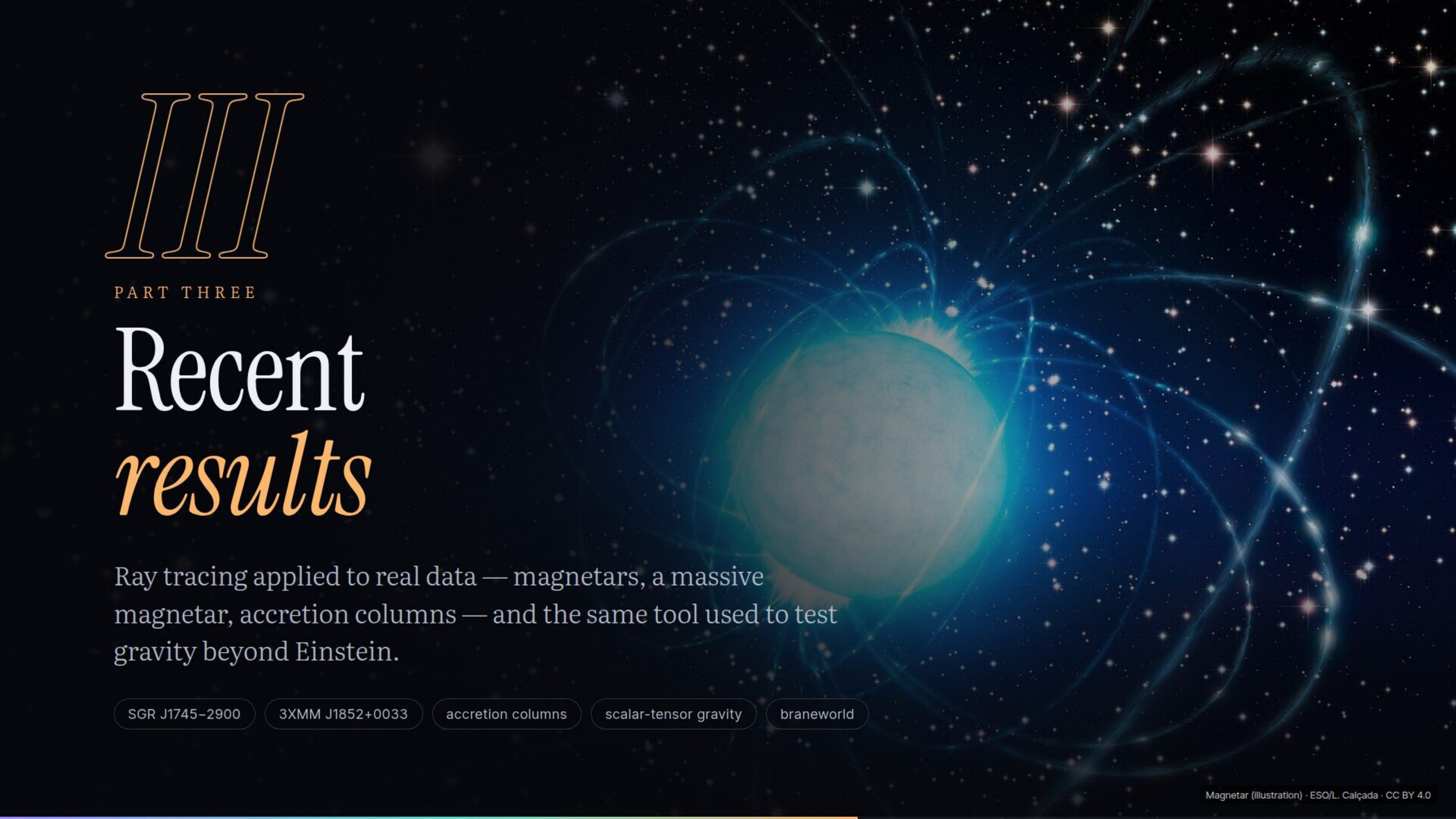
III

PART THREE

Recent *results*

Ray tracing applied to real data — magnetars, a massive magnetar, accretion columns — and the same tool used to test gravity beyond Einstein.

SGR J1745−2900 · 3XMM J1852+0033 · accretion columns · scalar-tensor gravity · braneworld



Magnetar (illustration) · ESO/L. Calçada · CC BY 4.0

2017 – 2026

Light curves *as an instrument*

2017 — **A&A 599, A87**: rotation-powered SGRs and AXPs, with realistic neutron stars

2020 — **ApJ 889, 165**: SGR J1745–2900 and a multipolar field, from the light curve

2023 — **Galaxies 11, 18**: pulse profiles of accretion columns

2024 — **JHEAp 42, 52**: 3XMM J1852+0033, a massive low-field magnetar

2024 — **EPJC 84, 1337**: light curves as a test of scalar-tensor gravity

2024 — **PRD 110, 103014**: stratified rotation and the moment of inertia

2025 — **PRD 112, 083010**: quark stars and the QCD equation of state

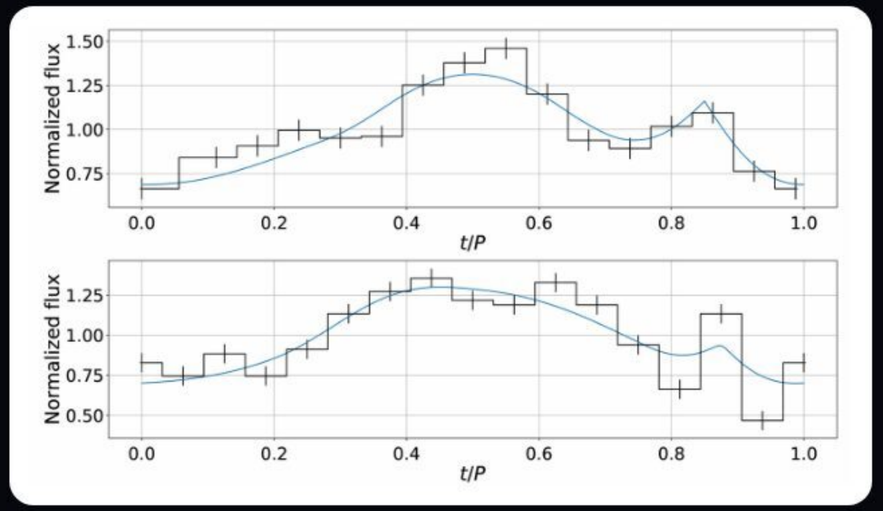
2026 — **arXiv:2607.09836**: compact stars in a high-tension braneworld

2026 — **JHEAp (submitted)**: the column of 1A 0535+262 through its NuSTAR spectrum

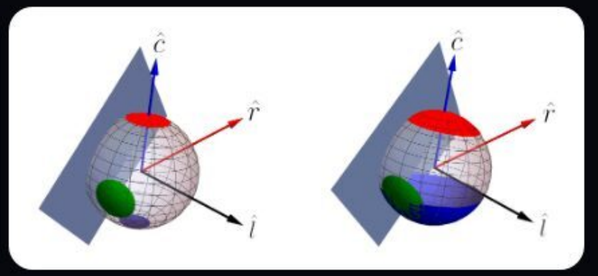
The common thread: reading in the light of a compact star its *mass*, its *radius*, its *field* — and the *gravity* that curves it.

DE LIMA, COELHO, PEREIRA, RODRIGUES & RUEDA · APJ 889, 165 (2020)

SGR J1745−2900: *more than two spots*



Chandra light curves (2013 and 2016) and the three-spot fits.

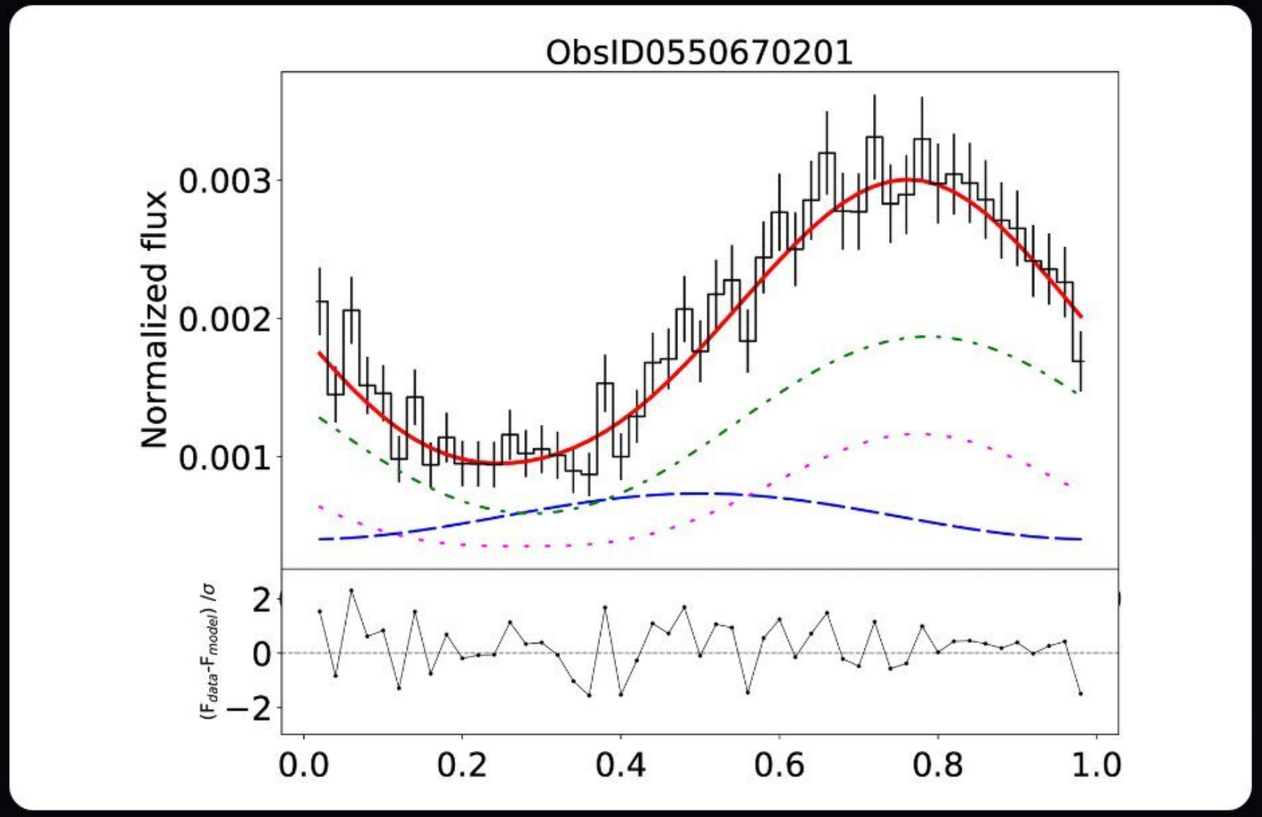


The three spots: two at the poles, one in the southern hemisphere.

- The magnetar 0.1 pc from Sgr A*; 3.5 years of **Chandra** data.
- Model: M, R, up to **three spots** of any size and position, Schwarzschild light bending; fitted with a genetic algorithm.
- Two or three spots describe the data; with three, they are **large** (16–67°).
- A field with non-trivial poloidal and toroidal structure — the same message NICER gave for J0030.

1.4 ± 0.1	10.9 ± 1.5
$M_\odot$	km

3XMM J1852+0033: *ray tracing and inference*



One of the five pulse profiles (XMM-Newton, 2008) and the best fit: dashed, the contribution of each spot.

- A **transient magnetar** near the remnant Kes 79, P = 11.56 s; until then there were only upper limits on the field.
- Five **XMM-Newton** epochs over three weeks in 2008.
- **Ray tracing** in Schwarzschild with circular spots of any size (Turolla & Nobili 2013) and **magnetised atmospheres** (NSMAX: H, C, O), with anisotropic emission.
- **Bayesian inference** (MCMC): mass, radius, field and spot positions shared by all epochs; size and temperature free to vary.
- Tested by **injection and recovery** of simulated data.

BEST MODEL

Three small spots, two in the southern hemisphere, and a **carbon** atmosphere: $\chi^2/\mathrm{dof} = 1.22$, well ahead of two spots or of hydrogen.

A *massive*, low-field magnetar

$2.09^{+0.16}_{-0.09}\ M_\odot$

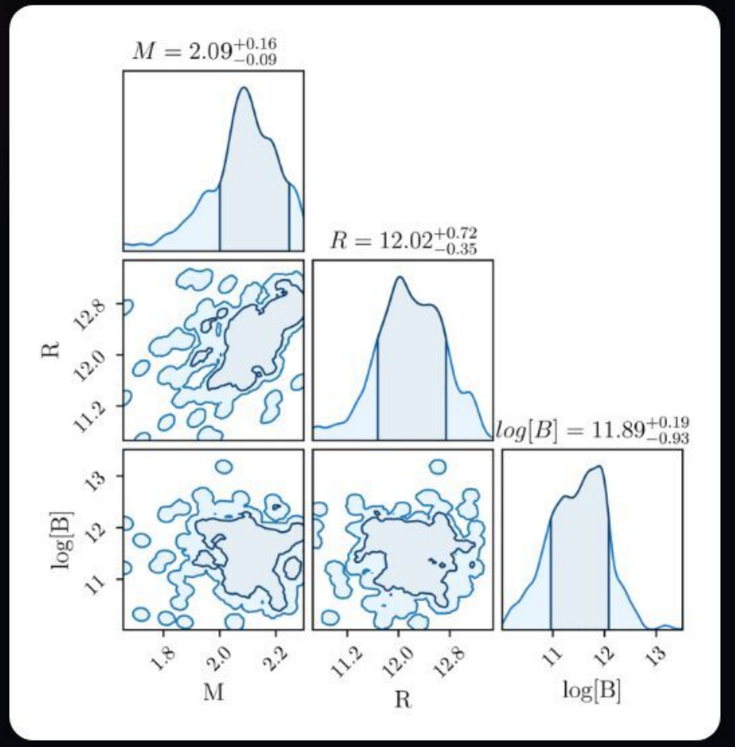
mass (1σ): among the highest ever estimated

12.0 ± 1.4 km

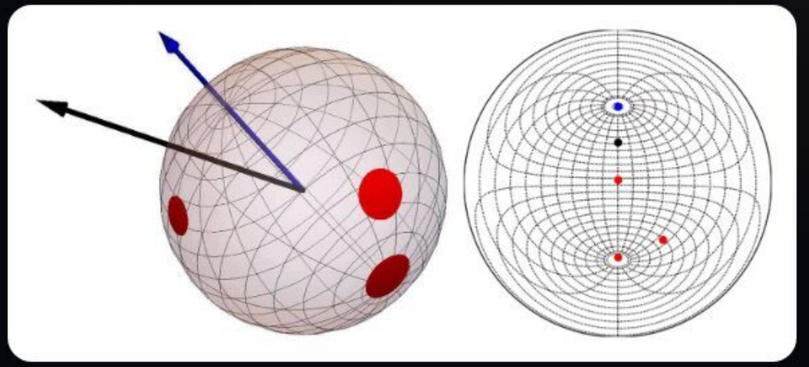
radius (2σ)

$\sim 8 \times 10^{11}$ G

surface field: log B = $11.89^{+0.19}_{-0.93}$, below the critical value



Posteriors of M, R and log B.

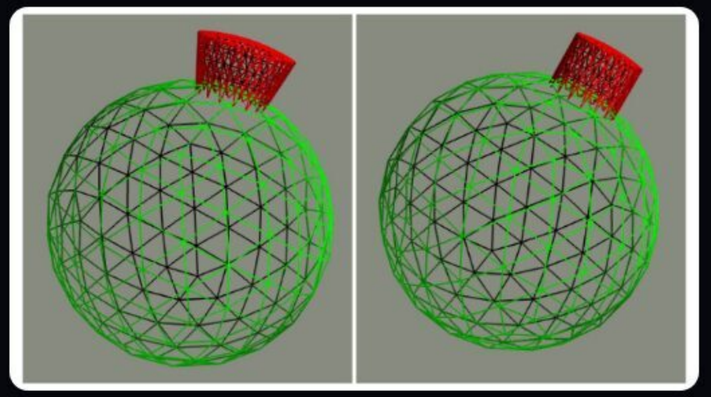


The best-fit geometry.

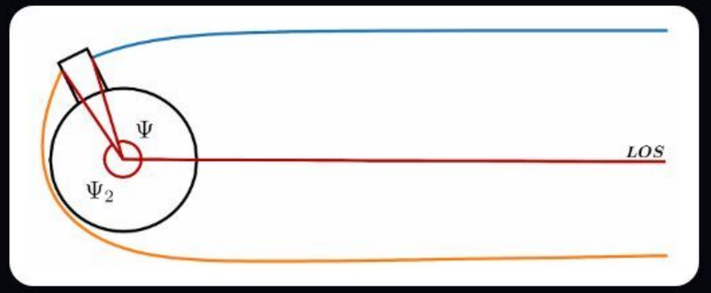
Interpretation: accretion of supernova material **buried** the field, which is now re-emerging; the carbon atmosphere would come from diffusive nuclear burning of H/He.

Compactness $GM/Rc^2 \approx 0.26$: favours equations of state of intermediate stiffness.

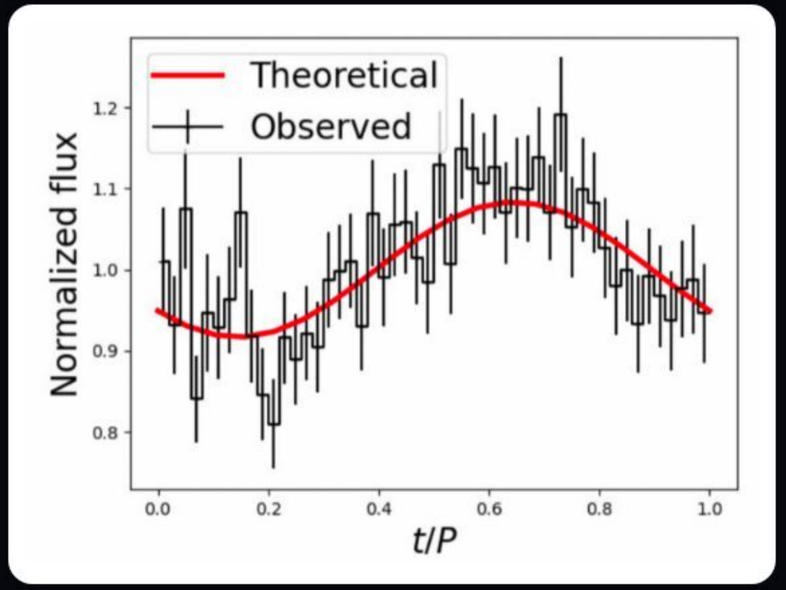
When the light comes *from a column*



The star and the columns (truncated cone and cylinder) as a 3D mesh.



Photons emitted above the surface may pass through a **periastron** before escaping.



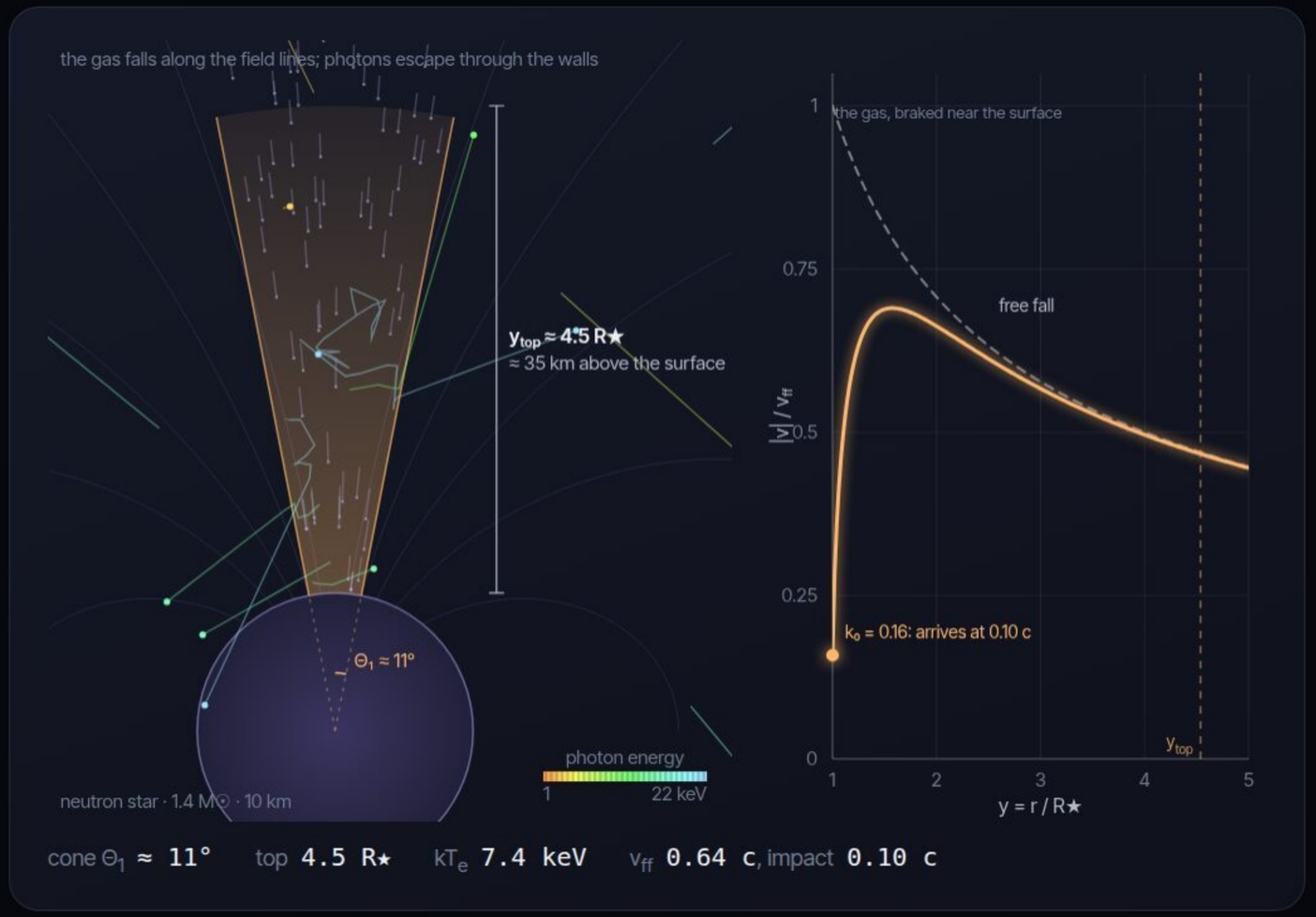
1A 0535+262 (NuSTAR, 2018): observed profile and the two-column model.

- In X-ray binaries, matter falls along the field lines and forms **columns** above the poles: the emission comes out sideways (fan beam).
- **Exact** ray tracing in Schwarzschild, tabulated, including for points above the surface; Doppler included.
- Genetic algorithm for the priors, then MCMC.

$1.53^{+0.52}_{-0.40}$ M⊙

$12.1^{+1.1}_{-0.9}$ km

Inside the column, *through the spectrum*

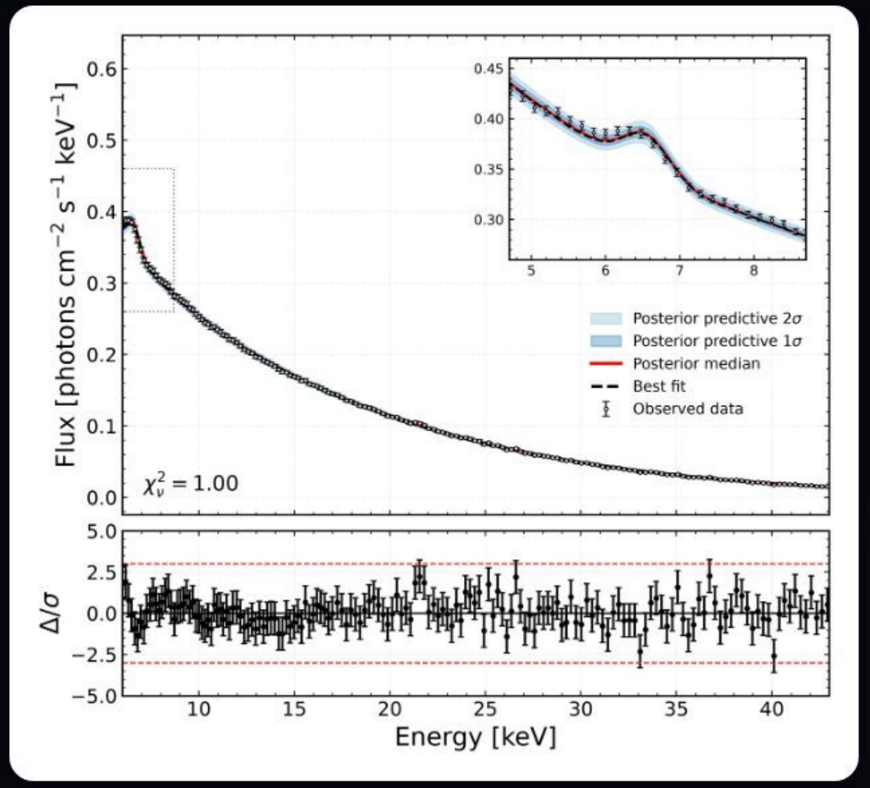


- The same star as on the previous slide, now through its **spectrum**: NuSTAR, 3–44 keV, L ≈ 3.7×10³⁶ erg/s — 16 times below the critical luminosity.
- **Becker & Wolff (2022)** model: a conical column; photons gain energy from the hot electrons and from the infalling gas (thermal and *bulk* Comptonization) and escape through the walls.
- 12 parameters, 4 independent MCMC chains of 256 walkers. The posterior has **two modes**; the dominant one wins by $\Delta \ln \mathscr{P} \approx 55$.

$$v(y) = -v_{ff}\left[\frac{k_\infty^2(y^3-1)+k_0^2}{y^4}\right]^{1/2}$$

y = r/R★ · FREE FALL HIGH UP, BRAKED NEAR THE SURFACE

A hot, tall column — *and a missing piece*



The NuSTAR spectrum and the posterior (median and 1σ and 2σ bands), with the residuals. Inset: the iron line.

8.56 ×10⁷ K

electron temperature, to 1% · kT ≈ 7.4 keV

4.5 R★

top of the column, ~35 km above the surface

0.070

ψ, bulk/thermal: **thermal** Comptonization dominates

$\log \theta_0 - 1.36 \log \dot{M} = -22.32 \pm 0.10$

the spectrum fixes the combination: more $\dot{M}$, a wider column

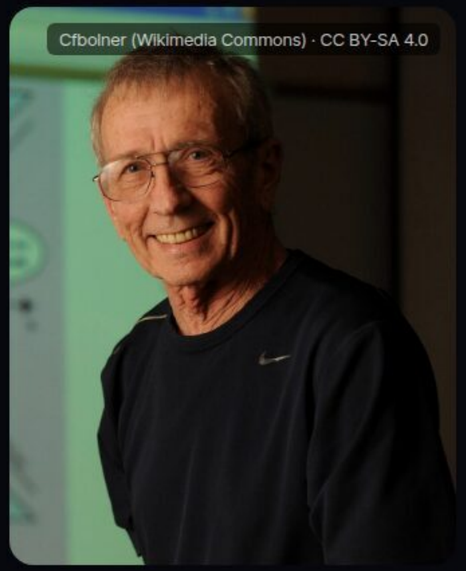
A MISSING PIECE

The field comes out 2.3σ above that of the cyclotron line — which would then sit at 55 keV, not the measured 46 — and the residuals move together (runs test: z = −3.7). Candidate: recombination between excited Landau levels, seen in this source and absent from the model.

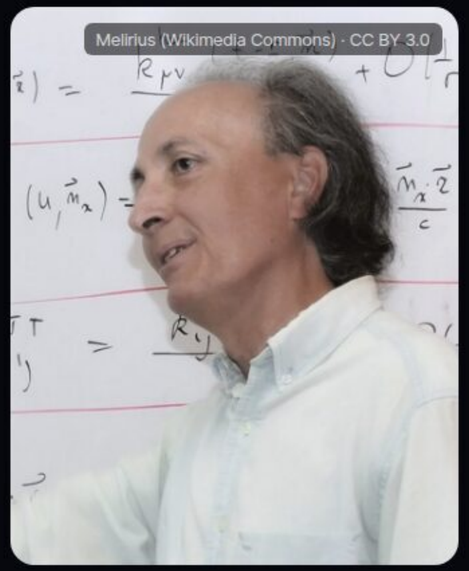
Dominant posterior mode. $\chi^2_\nu = 1.00$ with a free systematic uncertainty is not a goodness-of-fit test: the residuals are the diagnostic.

BRANS & DICKE 1961 · DAMOUR & ESPOSITO-FARÈSE 1993

What if gravity *is not Einstein's?*



Carl Brans
1935 – 2022



Thibault Damour
b. 1951

In **scalar-tensor** theories, besides the metric there is a scalar field ϕ that couples to matter. In the Solar System the effect is tiny; in a neutron star **spontaneous scalarisation** can occur: the field grows on its own.

Outside the star, the metric is no longer Schwarzschild: it is the **Just metric**, with a scalar charge Q:

$$ds^2 = -(1-\frac{a}{r})^{b/a}dt^2 + (1-\frac{a}{r})^{-b/a}dr^2 + (1-\frac{a}{r})^{1-b/a}r^2d\Omega^2$$

$B = 2GM/C^2 \cdot A/B = \sqrt{(1+Q^2)}$ · EINSTEIN FRAME

Two stars with the **same mass and the same radius** — "doppelgängers" — bend light in different ways. The light curve can tell the theories apart.

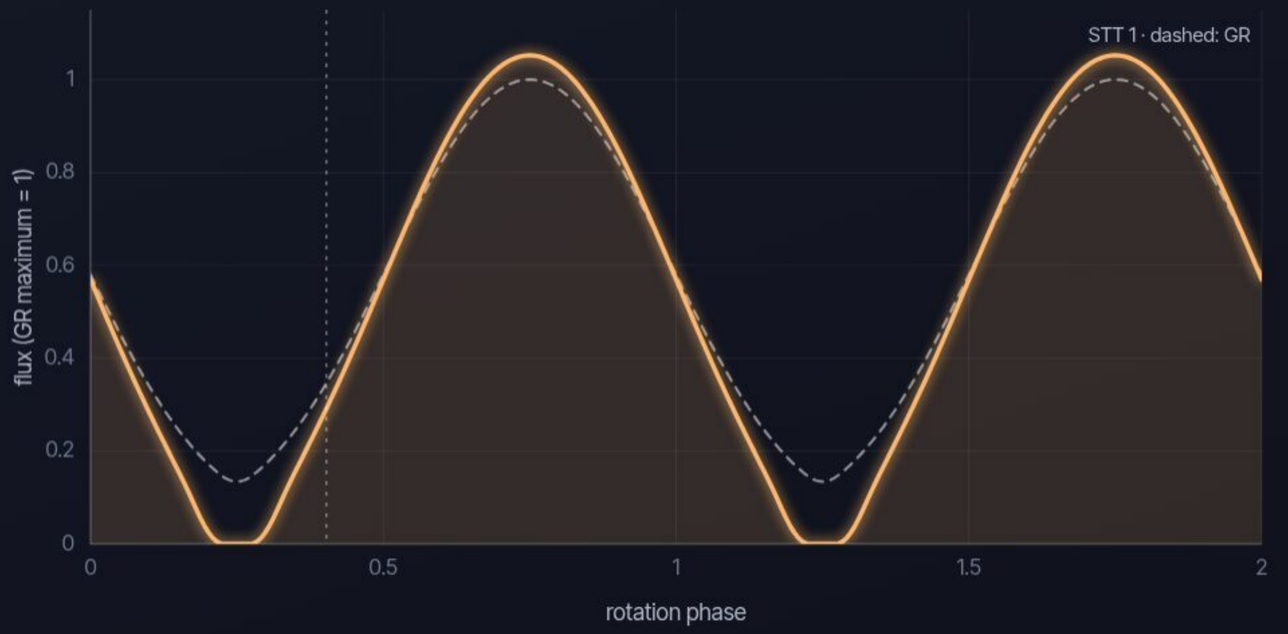
The same star, *different gravity*

M = 2.1 M⊙ · ONE 10° SPOT · ι = θ = 90° · STATIC STAR · EXACT INTEGRAL

GR | STT 1 · A_s = 0.95 | STT 2 · A_s = 1.05

the star seen from afar · dashed: its size in GR



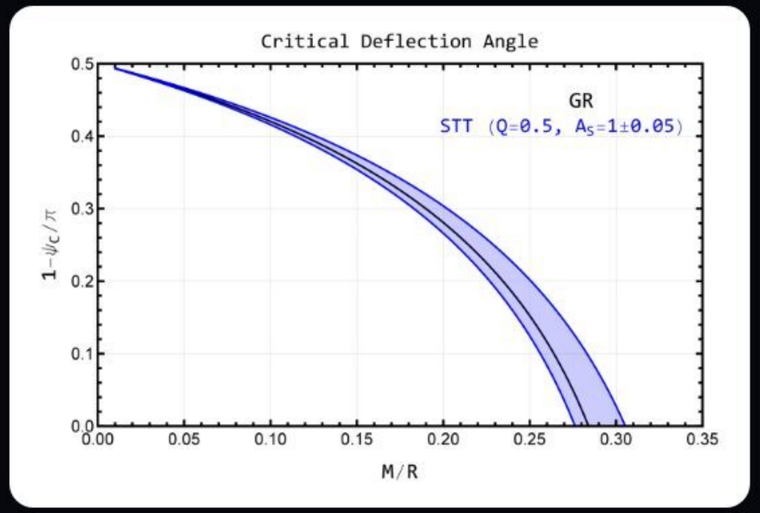


compactness, GM/Rc^2 0.284

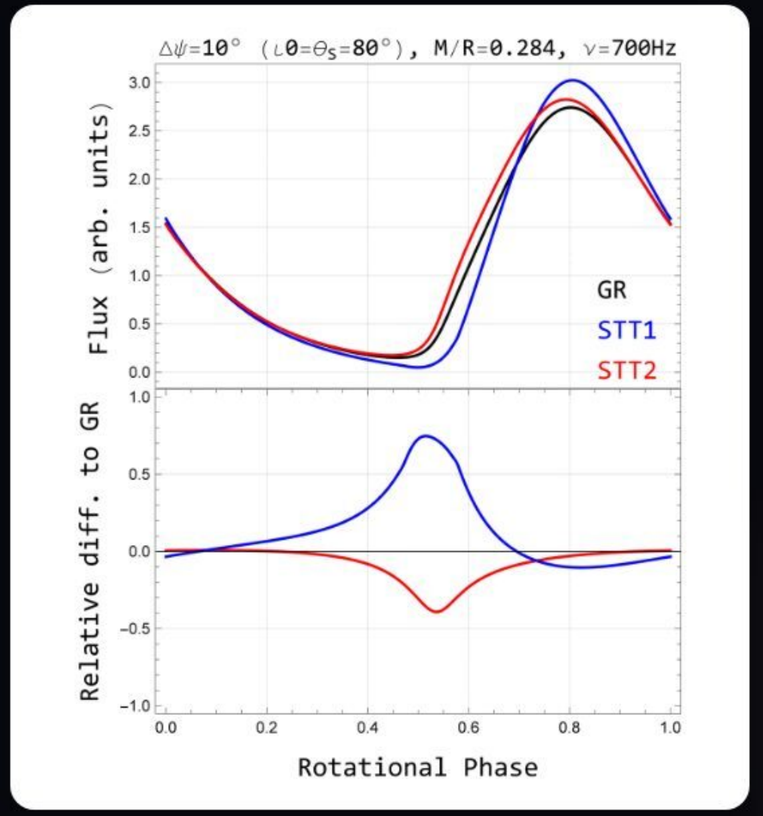
scalar charge, Q 0.50

STT 1: ψ_c 0.903π GR: ψ_c 1.000π invisible zone 17.4° R 10.92 km apparent radius 16.88 km PF 1.00

Up to *80%* difference in flux



The critical angle versus *M*/*R*: the GR line and the scalar-tensor band.



Spinning at 700 Hz, with Doppler and delays: flux and difference relative to GR.

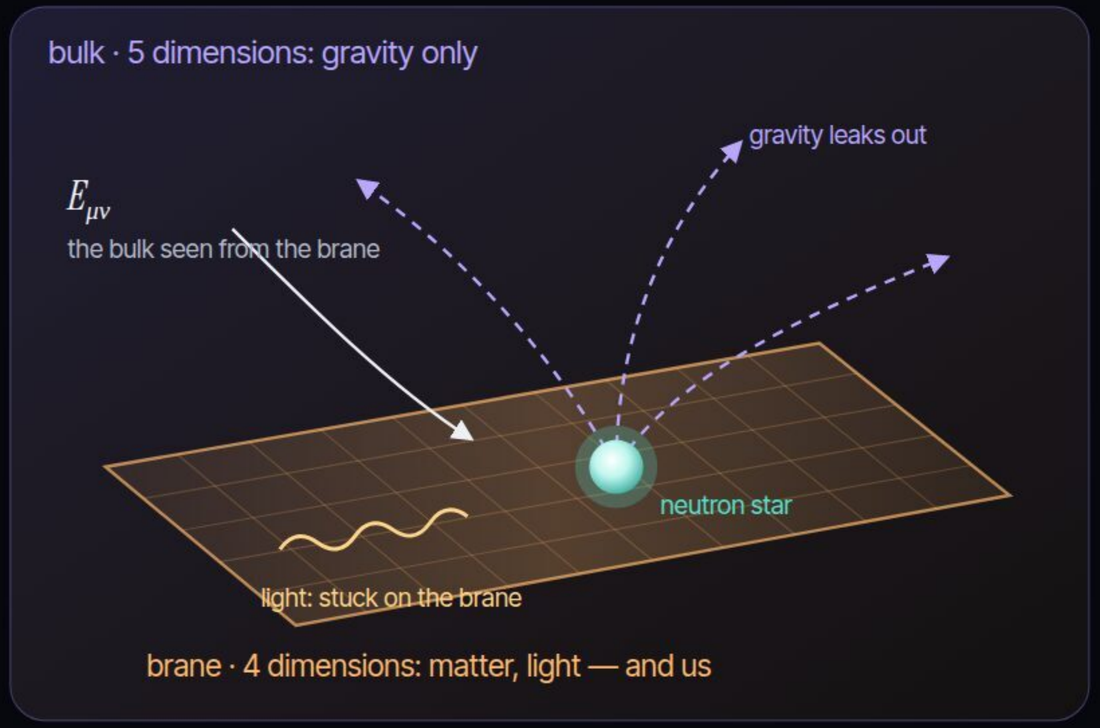
- The compactness at which $\psi_c = \pi$ ranges from **0.275 to 0.305** in scalar-tensor theory, against 0.284 in GR.
- The flux from a spot differs from GR by up to **~80%**.
- The radius changes by up to **~10%** for the same stellar parameters: today, this limits $Q \lesssim 0.5$.
- With NICER, J0740+6620 has $C = 0.21$–0.28: not yet enough to separate the theories.

OUTLOOK

Radii with smaller uncertainties — eXTP, NewAthena, and gravitational waves from Einstein Telescope and LISA — may constrain scalar-tensor theories through the **pulse shape**.

RANDALL & SUNDRUM 1999 · SHIROMIZU, MAEDA & SASAKI 2000

The Universe on a *membrane*



In a **braneworld**, our Universe is a four-dimensional hypersurface — the **brane** — inside a space with one extra dimension, the **bulk**. Matter and light are stuck on the brane; only gravity propagates in the bulk.

The idea was put forward to attack the hierarchy problem (why gravity is so weak) and the cosmological-constant problem and accelerated expansion.

$$G_{\mu\nu} = 8\pi T_{\mu\nu} + \kappa_5^4 S_{\mu\nu} - E_{\mu\nu}$$

EFFECTIVE EINSTEIN EQUATIONS ON THE BRANE · SHIROMIZU, MAEDA & SASAKI 2000

- $S_{\mu\nu}$: **local** corrections, quadratic in the matter ($\propto T^2$). They matter when the density approaches the **brane tension** λ.
- $E_{\mu\nu}$: the bulk **Weyl** tensor projected onto the brane — **nonlocal**: it cannot be computed from brane quantities alone.
- For $\lambda \to \infty$ and $E_{\mu\nu} \to 0$, General Relativity is recovered.

In the paper: an effective 4D closure, inspired by the Sahni–Shtanov scenario (2003) — not a complete 5D bulk solution.

Same nuclear physics, *different gravitation*

The equation of state of the matter (SLy) **does not change**. What changes is how much it **gravitates**: the star now feels an effective density and pressure.

$$\rho_{\text{eff}} = \rho + \frac{\rho^2}{2\lambda} + \rho_{\mathcal{W}}$$

$$p_{\text{eff}} = p + \frac{p\rho}{\lambda} + \frac{\rho^2}{2\lambda} + p_{\mathcal{W}}$$

LOCAL TERM (λ) · WEYL SECTOR ($\mathcal{W}$)

$$\frac{dm}{dr} = 4\pi r^2 \rho_{\text{eff}}$$

$$\frac{dp}{dr} = -(\rho + p)\,\frac{m + 4\pi r^3 p_{\text{eff}}}{r(r - 2m)}$$

MATTER IS CONSERVED: THE $(\rho + p)$ FACTOR IS THE USUAL ONE

The local term **strengthens attraction** and shrinks the star — but only when ρ approaches λ. At the centre of a neutron star $\rho \approx 2\times10^{-3}$ km^{-2}; with $\lambda \sim 10^{4}$ km^{-2}, $\rho/\lambda \sim 10^{-7}$.



WHAT COMES FROM THE BULK

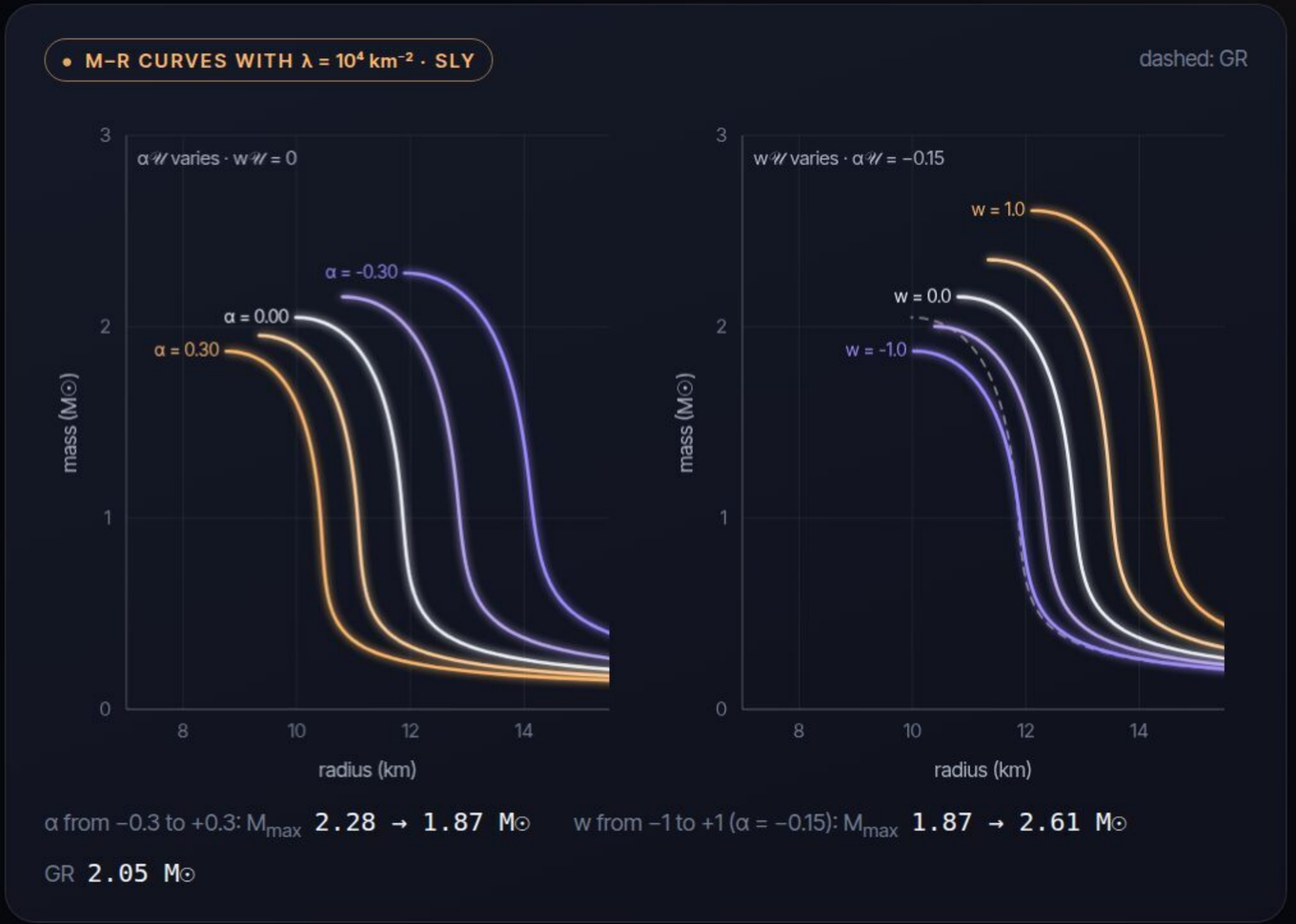
The Weyl sector: *screening* or extra attraction

$E_{\mu\nu}$ is the imprint of the bulk's gravitational field on the brane. There it acts as an **effective fluid** — in cosmology, the "dark radiation". But the brane does not determine this fluid: a **closure** is needed.

$$\rho_{\mathscr{U}} = \alpha_{\mathscr{U}}\,\rho \qquad p_{\mathscr{U}} = w_{\mathscr{U}}\,\rho_{\mathscr{U}}$$

Isotropic closure; the anisotropic Weyl stress ($\Delta_{\mathscr{U}}$) is left for future work.

IF…	EFFECT ON THE STAR
$\alpha_{\mathscr{U}} < 0$	$\rho_{\text{eff}} = \rho(1+\alpha_{\mathscr{U}}) < \rho$: **screening** — less mass gravitates, larger and heavier stars
$\alpha_{\mathscr{U}} > 0$	more attraction: smaller and lighter stars
$w_{\mathscr{U}}$	only appears as $w_{\mathscr{U}}\alpha_{\mathscr{U}}\rho$ in the effective pressure: it matters little when $\alpha_{\mathscr{U}}$ is small, and it can **compensate** $\alpha_{\mathscr{U}}$

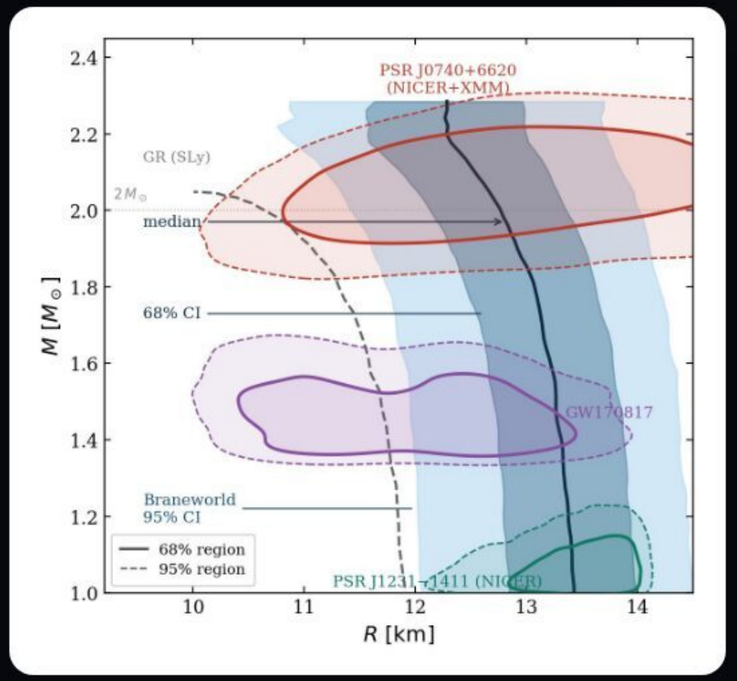


High tension, *mildly negative* Weyl coupling

DATA AND INFERENCE

GW170817 (radii from the tidal deformability) + NICER: PSR J0740+6620 and PSR J1231–1411 — both of them results of pulse-profile **ray tracing**. Ensemble MCMC (4 × 32 walkers × 3,000 steps, 288,000 samples), priors centred at large λ and at the GR values.

	MEDIAN	68%	GR
$\log_{10}(\lambda/\mathrm{km}^{-2})$	3.98	2.54 – 5.50	∞
$\alpha_{\mathcal{W}}$	−0.15	−0.23 – 0.15	0
$w_{\mathcal{W}}$	0.24	−0.78 – 0.61	0
$M_{\max}$ (M⊙)	2.30	2.22 – 2.44	2.05
$R_{1.4}$ (km)	13.31	12.74 – 13.85	11.76

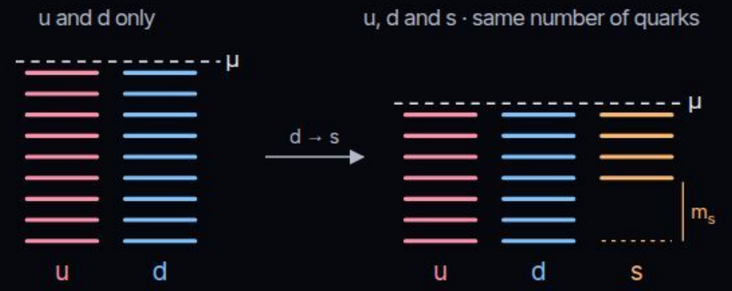


The posterior M–R band (68% and 95%), GR+SLy (dashed) and the constraints used.

- **$\lambda \sim 10^4$ km^{-2}**: the local term vanishes ($\rho/\lambda \sim 10^{-7}$). The posterior nearly repeats the prior: the data are *compatible* with high tension, they do not measure λ.
- **$\alpha_{\mathcal{W}} < 0$ in 83% of the samples**: a mild screening. It is the Weyl sector, not the local term, that pushes the curve to the right.
- **A secondary branch (17%)**: $\alpha_{\mathcal{W}} > 0$ and $w_{\mathcal{W}} < 0$ compensate — smaller $R_{1.4}$, larger $M_{\max}$, compactness near the imposed bound.
- **$M_{\max} \approx 2.3$ M⊙**: consistent with PSR J0952–0607 (2.35 ± 0.11 M⊙). GW190814 (2.50–2.67 M⊙) only fits in the 95% tail.
- **The message**: heavier stars are not a generic prediction of braneworlds — they depend on the **sign of the Weyl sector**.

What if iron is *not* the bottom of the well?

At extreme densities neutrons may dissolve into quarks. The Bodmer–Witten hypothesis goes further: matter made of **u, d and s** quarks would be the **ground state** of matter — with an energy per baryon below that of ^{56}Fe, 930 MeV.



Pauli: each flavor has its own Fermi sea. A third flavor opens new levels and the top (μ) drops — even after paying for the s mass.

- Nuclei don't decay because that would take ~A **simultaneous** weak d → s conversions.
- And u–d matter alone must be **unstable** — otherwise iron itself would have turned into quarks.
- If it holds, "neutron stars" may be **quark stars**, bound by the strong force.



Dense QCD in *three numbers*

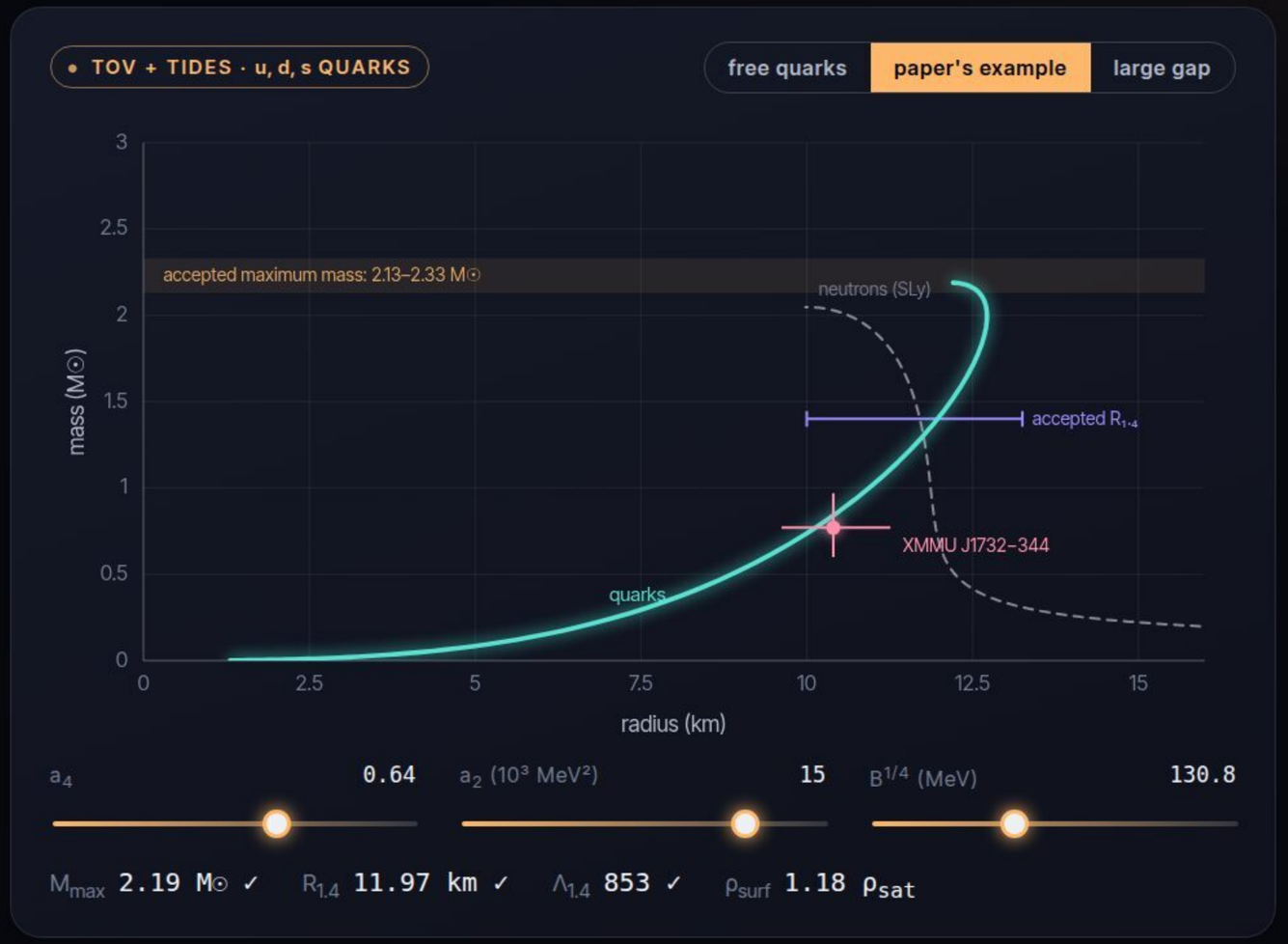
$$\omega = -\frac{3a_4}{4\pi^2}\mu^4 + \frac{3a_2}{4\pi^2}\mu^2 + B$$

THERMODYNAMIC POTENTIAL · $p = -\omega$ · μ: QUARK CHEMICAL POTENTIAL

$a_4 = 1 - c$	the **strong interaction**. $a_4 = 1$ means free quarks; perturbative QCD suggests $c \gtrsim 0.3$
$a_2 = m_s^2 - 4\Delta^2$	the s-quark mass against the **gap** Δ of color superconductivity (CFL phase) — it can be negative
B	the **bag constant**: the cost of opening the QCD vacuum

At the surface $p = 0$, but the density is not: $\rho_s \approx 4B \sim 10^{14}$–$10^{15}$ g/cm^3. The star is **self-bound**: at small masses $M \propto R^3$, unlike a neutron star.

TOV + the tidal equation with the surface jump ($\Delta y = -4\pi R^3 \rho_s / M$), solved here. The paper's example gives 2.19 M⊙, 11.97 km and $\Lambda = 852$ — the panel reproduces it.

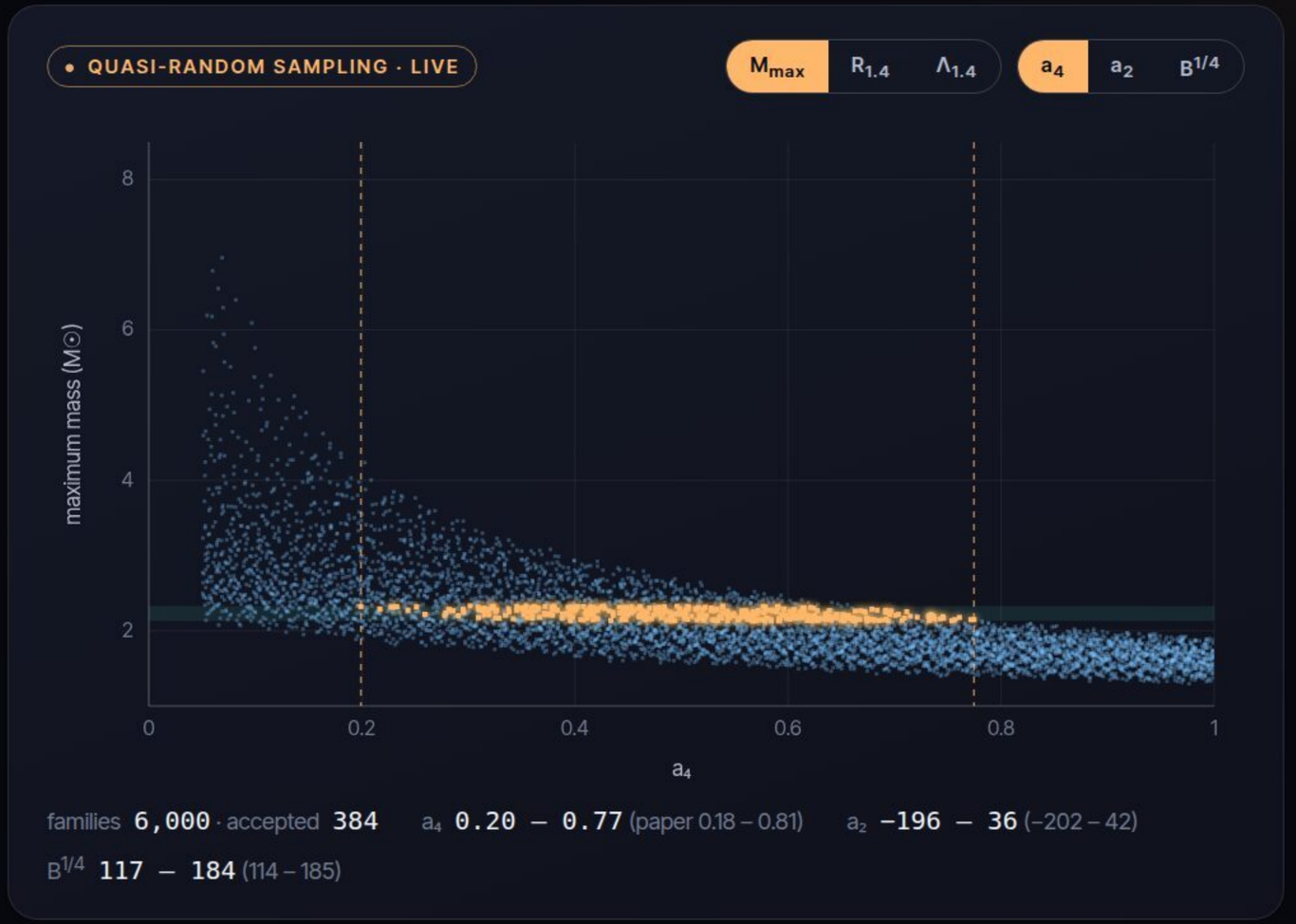


Thousands of stars, *drawn at random*

Each point (a_2, a_4, B) inside the stability window is a **whole family** of stars: TOV for M and R, the tidal equation for Λ. The paper drew **263,287** points by Latin hypercube sampling; here the browser draws its own, live.

	ACCEPTED RANGE	WHERE IT COMES FROM
M_{max}	2.13 – 2.33 M⊙	pulsars in binaries (Romani et al. 2022) · GW170817 + GRB 170817A (Rezzolla et al. 2018)
$R_{1.4}$	10.00 – 13.25 km	NICER radii — **ray tracing** — and gravitational waves (compilation by Pang et al. 2023)
$\Lambda_{1.4}$	< 970	tidal deformability from GW170817

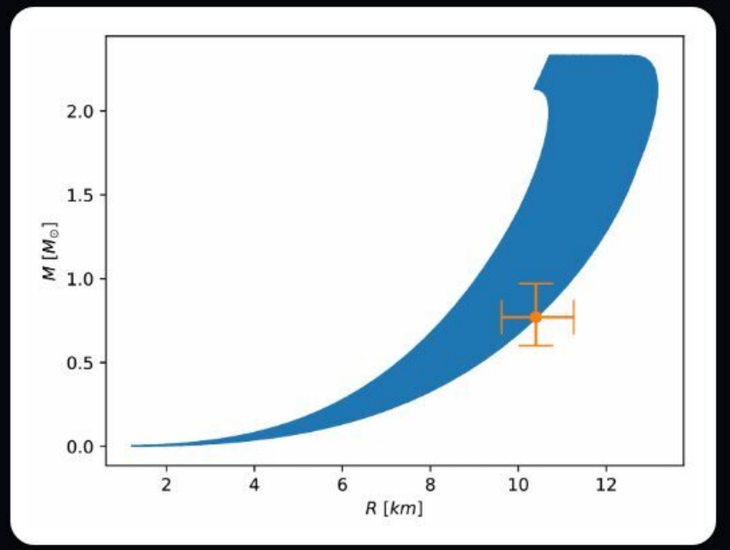
Amber: the families that pass all three tests. The dashed lines show how far the accepted ones reach — and they converge to the paper's limits.



Interacting quarks, a dense surface

COMBINED CONSTRAINTS	RANGE
a_4	0.18 – 0.81
a_2 (10^3 MeV^2)	−202 – 42
$B^{1/4}$ (MeV)	114 – 185
$R_{1.4}$ (km)	10.00 – 12.26
$\Lambda_{1.4}$	270 – 970
$\rho_{surface}$ (10^{15} g/cm^3)	0.29 – 0.60
c_s/c	0.51 – 0.85

The sound speed exceeds $c/\sqrt{3} \approx 0.58$ — the conformal limit of QCD at infinite density — when $a_2 < 0$.



All M–R curves that pass the constraints, with XMMU J173203.3−344518 (orange).

- $a_4 \lesssim 0.81$: the matter departs by at least **20%** from free quarks. Strong interactions cannot be neglected, at any mass.
- **$\Delta \lesssim 230$ MeV** (with $m_s = 100$ MeV), from the lower limit on a_2 — in line with the independent bound of $\lesssim 216$ MeV (Kurkela et al. 2024).
- **The surface jumps from vacuum to 1.1–2.2 ρ_{sat}**: a number that enters the tides, the oscillation modes and the star's mountains.
- **$R_{1.4}$ = 10.0–12.3 km**, like a neutron star: the radius alone cannot tell them apart; the tides can.
- **XMMU J1732−344** (0.77 M⊙, 10.4 km, in HESS J1731−347) falls on the curves effortlessly — a quark-star candidate, if the atmosphere model behind its radius is right.

When core and crust *do not spin together*

Almost every calculation treats a pulsar as a **rigid body**. But the interior may be superfluid, have a phase transition — a quark core — or gain accreted layers that spin differently, like the Sun's tachocline.

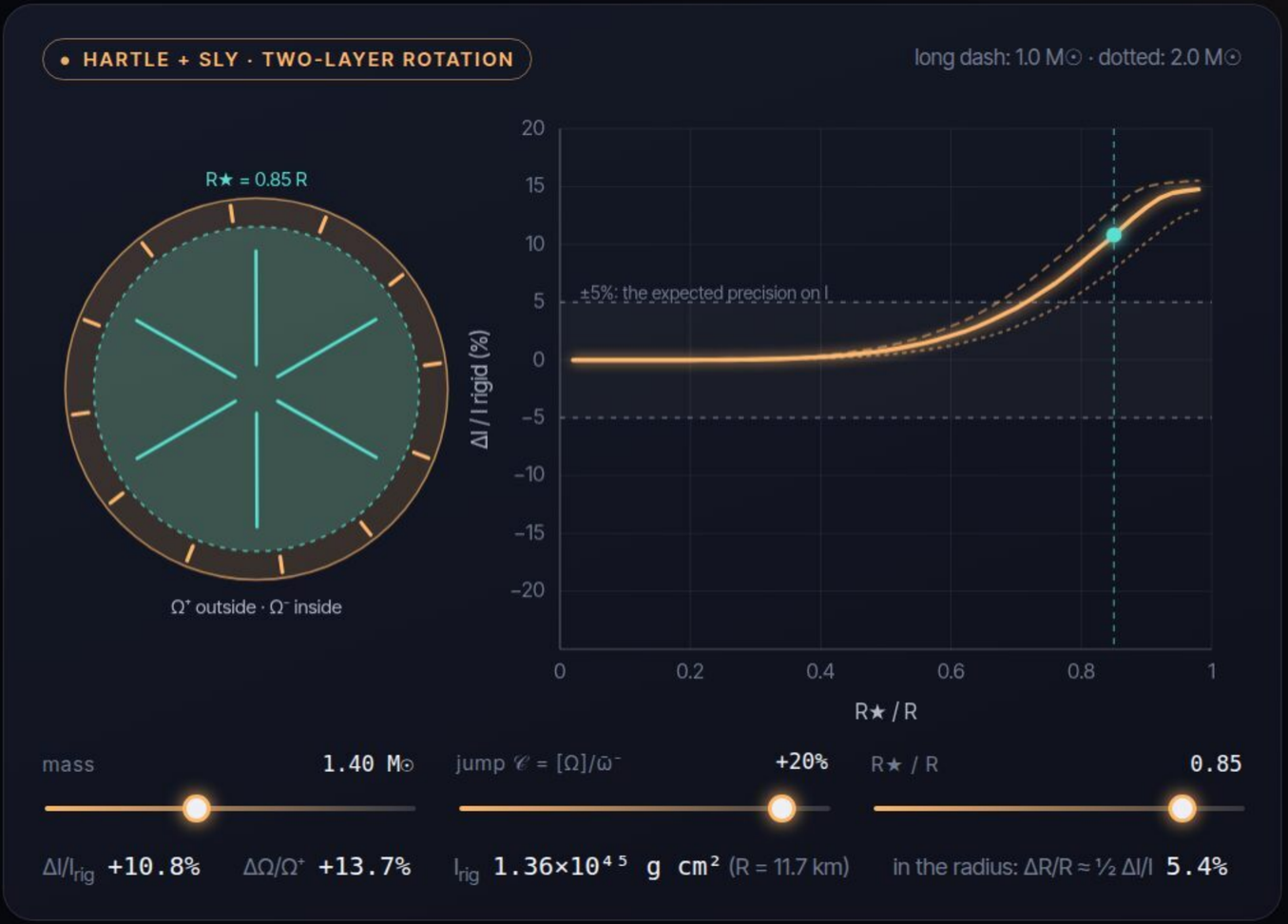
$$\frac{1}{r^4}\frac{d}{dr}(r^4 j\,\bar{\omega}') + \frac{4j}{r}\bar{\omega} = 0$$

FRAME DRAGGING IN SLOW ROTATION · HARTLE 1967 · $\bar{\omega} = \Omega - \omega$

$$[\bar{\omega}]_{R\star} = [\Omega],\ [\bar{\omega}']_{R\star} = 0 \cdot I = \frac{J}{\Omega^+}$$

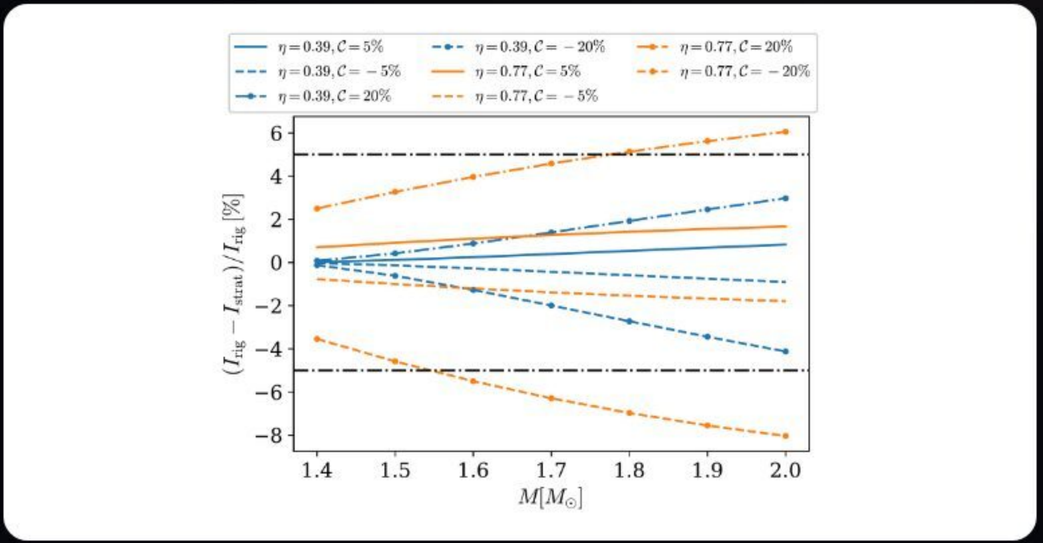
THE JUMP AT R★, DERIVED IN THE PAPER · Ω^+: THE SURFACE SPIN, THE ONE TIMING MEASURES

J comes from the derivative of $\bar{\omega}$ at the surface, so it **feels the jump**: the same star, with the same Ω^+, has a different moment of inertia.



Rigid rotation is an *assumption* — and it has a cost

- **Near the surface, $\Delta I/I \to \Delta\Omega/\Omega$**[+]; below ~40% of the radius the effect vanishes. It grows fast for $R\star \gtrsim 0.6$ R, and more in light stars.
- **Hybrid stars**: the jump sits at the transition to the quark core, and only exceeds 5% with $\mathscr{C}$ ~ 20% and $R\star \gtrsim 0.8$ R (figure).
- **In the radius**: with a known mass, $\Delta R/R = \frac{1}{2}\,\Delta I/I$. Ten percent in I is 5% in R — the size of the NICER error bars, which come from **ray tracing**.
- **Direct measurement of I**: the double pulsar J0737–3039A should give I to 11% by 2030 (today, I_{45} = 1.15–1.48). Jumps of 10–20% in Ω cannot be told from rigid rotation.
- **Energy**: with I 10–20% larger, five Fermi gamma-ray pulsars with $L_\gamma > \dot{E}$ fit back within the rotational energy.
- **I-Love-Q**: the quadrupole depends on another function besides J — the universal relations weaken. And in **CCOs**, accreted layers may spin differently for up to ~10 thousand years.



Hybrid stars (η = 0.39 and 0.77): Δ//I with the jump at the phase transition, against mass.

PULSAR (FERMI)	$L_\gamma/\dot{E}$, I_0	+10%	+20%
J1057–5851	1.19	> 1	0.99
J1522–5735	1.09	0.99	0.91
J1650–4601	1.11	> 1	0.92

CONCLUSIONS

From geometry *to observation* — and back

- In a neutron star, light bending, redshift and delays are **not corrections**: they define the image and the light curve.
- Backward ray tracing — with the exact integral when the compactness calls for it — links the metric to what a telescope measures.
- Applied to Chandra, XMM-Newton and NuSTAR data, it gave the mass, radius, field and spot geometry of magnetars and of an accreting star — and, through its spectrum, the temperature and height of its column.
- Since the light curve depends only on the exterior metric, it also **tests gravity**: scalar-tensor theories and braneworld.
- And the interior matters: **quark stars** and **stratified rotation** change the M, R and I that ray tracing measures.

NEXT STEPS

- Magnetised atmospheres and exact ray tracing together, for isolated cooling stars.
- Fast rotation and metrics beyond Schwarzschild in the same fits.
- Data from the next generation: eXTP, NewAthena, STROBE-X.

The panels in this seminar run in the browser: *rafael-lima.pages.dev/talks*.

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Simulated Sobral plate — positions from the Tycho-2 catalogue (Høg et al., A&A 355, L27, 2000), via VizieR/CDS

P–Ṗ diagram — ATNF pulsar catalogue 2.8.1 (Manchester et al., AJ 129, 1993, 2005)

THANK YOU

Questions?

Slides, animations and dashboards available at
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